

Management Guide



Hy-Line®

W-36

USE OF THE MANAGEMENT GUIDE

The genetic potential of Hy-Line W-36 Commercial can only be realized if good poultry husbandry practices and management are used. This management guide outlines successful flock management programs for Hy-Line Variety W-36 Commercial based on field experience compiled by Hy-Line International and using an extensive commercial layer flock database of Hy-Line flocks from all parts of the world. Hy-Line International Management Guides are periodically updated as new performance data and/or nutrition information become available.

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Summary of Performance Standards

GROWING PERIOD (TO 17 WEEKS):	
Livability	97.5%
Feed Consumed	11.8–13.1 lb
Body Weight at 17 Weeks	2.72–2.80 lb
LAYING PERIOD (TO 100 WEEKS):	
Percent Peak	95–97%
Hen-Day Eggs to 60 Weeks	256–264
Hen-Day Eggs to 90 Weeks	422–436
Hen-Day Eggs to 100 Weeks	471–487
Hen-Housed Eggs to 60 Weeks	252–260
Hen-Housed Eggs to 90 Weeks	411–424
Hen-Housed Eggs to 100 Weeks	456–472
Livability to 60 Weeks	97.1%
Livability to 100 Weeks	92.0%
Days to 50% Production (from hatch)	143 days
Average Egg Weight at 26 Weeks	45.0 lb / case (56.6 g / egg)
Average Egg Weight at 32 Weeks	47.0 lb / case (59.2 g / egg)
Average Egg Weight at 70 Weeks	49.8 lb / case (62.8 g / egg)
Average Egg Weight at 100 Weeks	50.1 lb / case (63.1 g / egg)
Total Egg Mass per Hen-Housed (18–100 weeks)	62.4 lb (28.93 kg)
Body Weight at 26 Weeks	3.26–3.40 lb (1.48–1.54 kg)
Body Weight at 32 Weeks	3.30–3.43 lb (1.50–1.55 kg)
Body Weight at 70 Weeks	3.32–3.45 lb (1.51–1.56 kg)
Body Weight at 100 Weeks	3.32–3.45 lb (1.51–1.56 kg)
Freedom From Egg Inclusions	Excellent
Shell Strength	Excellent
Haugh Units at 38 Weeks	91.4
Haugh Units at 56 Weeks	87.5
Haugh Units at 70 Weeks	86.0
Haugh Units at 100 Weeks	83.0
Average Daily Feed Consumption (18–100 weeks)	22 lb / day per 100 birds (100 g / day per bird)
Feed Conversion Rate, lb feed / lb Eggs (20–60 weeks)	1.81–1.95
Feed Conversion Rate, lb feed / lb Eggs (20–100 weeks)	1.95–2.11
Feed Utilization, lb Egg / lb Feed (20–60 weeks)	0.53–0.55
Feed Utilization, lb Egg / lb Feed (20–100 weeks)	0.51–0.54
Feed Consumption per Dozen Eggs (20–60 weeks)	2.80–3.07 lb (1.27–1.39 kg)
Feed Consumption per Dozen Eggs (20–100 weeks)	3.10–3.40 lb (1.41–1.54 kg)
Condition of Droppings	Dry

Performance Summary data is based on results obtained from customers around the world.
Please send your results to info@hyline.com. An easy to use record-keeping program,
Hy-Line International EggCel, can be found at www.hylineeggcel.com.



Performance Tables

Rearing Period

AGE (weeks)	MORTALITY Cumulative (%)	BODY WEIGHT (kg)	BODY WEIGHT (lb)	FEED INTAKE (lb/day/100)	CUMULATIVE FEED INTAKE (lb/bird to date)	WATER CONSUMPTION (gal/day/100)	UNIFORMITY (Cage)
1	0.75	0.06 – 0.07	0.13 – 0.15	2.89–3.50	0.20 – 0.25	0.52 – 0.84	>85%
2	1.30	0.11 – 0.12	0.25 – 0.26	3.68–4.48	0.46 – 0.56	0.66 – 1.07	
3	1.55	0.18 – 0.18	0.39 – 0.41	4.78–5.83	0.79 – 0.97	0.86 – 1.40	
4	1.66	0.25 – 0.26	0.54 – 0.56	6.29–7.39	1.23 – 1.48	1.13 – 1.77	>80%
5	1.77	0.32 – 0.34	0.71 – 0.74	7.50–8.40	1.76 – 2.07	1.35 – 2.01	
6	1.88	0.41 – 0.43	0.91 – 0.94	8.70–9.53	2.37 – 2.74	1.56 – 2.28	
7	1.99	0.50 – 0.52	1.11 – 1.14	9.40–10.30	3.03 – 3.46	1.69 – 2.47	
8	2.10	0.59 – 0.61	1.30 – 1.34	10.13–11.26	3.74 – 4.25	1.82 – 2.70	
9	2.15	0.68 – 0.70	1.50 – 1.54	11.11–12.08	4.51 – 5.09	2.00 – 2.90	
10	2.20	0.78 – 0.80	1.72 – 1.77	11.82–12.81	5.34 – 5.99	2.12 – 3.07	
11	2.25	0.87 – 0.89	1.91 – 1.97	12.19–13.15	6.19 – 6.91	2.19 – 3.15	
12	2.30	0.95 – 0.97	2.09 – 2.15	12.43–13.51	7.06 – 7.86	2.23 – 3.24	>85%
13	2.35	1.02 – 1.04	2.24 – 2.30	12.86–14.15	7.96 – 8.85	2.31 – 3.39	
14	2.40	1.09 – 1.12	2.39 – 2.46	13.06–14.56	8.88 – 9.87	2.35 – 3.49	
15	2.45	1.15 – 1.19	2.54 – 2.61	13.41–14.90	9.82 – 10.91	2.41 – 3.57	
16	2.50	1.19 – 1.23	2.63 – 2.71	14.00–15.20	10.80 – 11.97	2.52 – 3.64	>90%
17	2.55	1.23 – 1.27	2.72 – 2.80	14.75–15.90	11.83 – 13.09	2.65 – 3.81	

Laying Period

AGE (weeks)	% HEN- DAY Current	HEN-DAY EGGS Cumulative	HEN- HOUSED EGGS Cumulative	MORT- ALITY Cumulative (%)	BODY WEIGHT (kg)	BODY WEIGHT (lb)	FEED INTAKE (lb/day/100)	WATER CONS. ¹ (gal/day/100)	HEN- HOUSED EGG MASS Cumulative (lb)	AVG. EGG WEIGHT ² (lb/case)
18	2 – 3	0.1 – 0.2	0.1 – 0.2	0.1	1.25–1.32	2.76– 2.90	14.0 – 16.4	2.77 – 4.02	0.02	35.4
19	15 – 22	1.2 – 1.8	1.2 – 1.7	0.1	1.27–1.37	2.80– 3.01	15.0 – 17.0	2.89 – 4.17	0.17	36.8
20	35 – 50	3.6 – 5.3	3.6 – 5.2	0.1	1.32–1.43	2.91– 3.15	16.0 – 17.7	3.01 – 4.33	0.48	38.0
21	60 – 67	7.8 – 9.9	7.8 – 9.9	0.2	1.38–1.47	3.04– 3.24	16.8 – 18.6	3.05 – 4.39	0.97	39.6
22	80 – 84	13.4 – 15.8	13.4 – 15.8	0.2	1.41–1.49	3.11– 3.28	17.7 – 19.5	3.17 – 4.54	1.63	41.3
23	88 – 92	19.6 – 22.3	19.6 – 22.2	0.3	1.43–1.50	3.15– 3.31	18.8 – 20.5	3.33 – 4.76	2.37	42.6
24	91 – 94	26.0 – 28.9	25.9 – 28.8	0.4	1.45–1.51	3.20– 3.33	19.7 – 21.2	3.45 – 4.91	3.15	43.6
25	93 – 95	32.5 – 35.5	32.4 – 35.4	0.5	1.47–1.53	3.24– 3.38	20.4 – 21.7	3.53 – 5.02	3.96	44.4
26	94 – 96	39.1 – 42.2	39.0 – 42.1	0.5	1.48–1.54	3.26– 3.40	20.8 – 22.2	3.61 – 5.12	4.79	45.0
27	95 – 96	45.7 – 49.0	45.6 – 48.8	0.6	1.48–1.55	3.27– 3.41	21.1 – 22.6	3.61 – 5.12	5.63	45.5
28	95 – 96	52.4 – 55.7	52.1 – 55.5	0.6	1.49–1.55	3.28– 3.42	21.2 – 22.6	3.61 – 5.12	6.48	45.9
29	95 – 97	59.0 – 62.5	58.7 – 62.2	0.7	1.49–1.55	3.29– 3.42	21.3 – 22.7	3.61 – 5.12	7.33	46.2
30	95 – 97	65.6 – 69.2	65.3 – 68.9	0.7	1.49–1.55	3.30– 3.42	21.4 – 22.7	3.61 – 5.12	8.19	46.5
31	95 – 97	72.3 – 76.0	71.9 – 75.6	0.8	1.49–1.55	3.30– 3.42	21.5 – 22.8	3.69 – 5.23	9.05	46.8
32	95 – 96	78.9 – 82.7	78.5 – 82.3	0.9	1.50–1.56	3.30– 3.43	21.5 – 22.8	3.69 – 5.23	9.92	47.0
33	94 – 96	85.5 – 89.5	85.0 – 89.0	0.9	1.50–1.56	3.31– 3.43	21.6 – 22.9	3.72 – 5.28	10.78	47.2
34	94 – 96	92.1 – 96.2	91.5 – 95.6	1.0	1.50–1.56	3.31– 3.44	21.6 – 22.9	3.72 – 5.28	11.65	47.4
35	94 – 95	98.7 – 102.9	98.1 – 102.2	1.0	1.50–1.56	3.31– 3.44	21.6 – 23.0	3.72 – 5.28	12.52	47.5
36	94 – 95	105.2 – 109.5	104.6 – 108.8	1.1	1.50–1.56	3.31– 3.45	21.6 – 23.1	3.72 – 5.28	13.38	47.7
37	93 – 95	111.8 – 116.2	111.0 – 115.4	1.1	1.51–1.56	3.32– 3.45	21.6 – 23.1	3.72 – 5.28	14.25	47.8
38	93 – 95	118.3 – 122.8	117.5 – 122.0	1.2	1.51–1.56	3.32– 3.45	21.7 – 23.1	3.72 – 5.28	15.11	47.9
39	93 – 95	124.8 – 129.5	123.9 – 128.5	1.3	1.51–1.56	3.32– 3.45	21.7 – 23.1	3.76 – 5.34	15.98	48.0
40	93 – 94	131.3 – 136.1	130.3 – 135.0	1.3	1.51–1.56	3.32– 3.45	21.7 – 23.1	3.76 – 5.34	16.84	48.1
41	92 – 94	137.7 – 142.7	136.6 – 141.5	1.4	1.51–1.56	3.32– 3.45	21.7 – 23.1	3.72 – 5.28	17.70	48.2
42	92 – 94	144.2 – 149.2	143.0 – 148.0	1.5	1.51–1.56	3.32– 3.45	21.7 – 23.1	3.76 – 5.34	18.56	48.3
43	92 – 93	150.6 – 155.8	149.3 – 154.4	1.6	1.51–1.56	3.32– 3.45	21.8 – 23.1	3.76 – 5.34	19.42	48.3
44	91 – 93	156.9 – 162.3	155.5 – 160.9	1.6	1.51–1.56	3.32– 3.45	21.8 – 23.1	3.76 – 5.34	20.27	48.4
45	91 – 93	163.3 – 168.8	161.8 – 167.2	1.7	1.51–1.56	3.32– 3.45	21.8 – 23.1	3.76 – 5.34	21.12	48.5
46	90 – 92	169.6 – 175.2	168.0 – 173.6	1.8	1.51–1.56	3.32– 3.45	21.8 – 23.1	3.80 – 5.39	21.97	48.6
47	90 – 92	175.9 – 181.7	174.2 – 179.9	1.9	1.51–1.56	3.32– 3.45	21.8 – 23.1	3.80 – 5.39	22.81	48.7
48	90 – 92	182.2 – 188.1	180.4 – 186.2	1.9	1.51–1.56	3.32– 3.45	21.8 – 23.1	3.80 – 5.39	23.66	48.7

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AGE (weeks)	% HEN- DAY Current	HEN-DAY EGGS Cumulative	HEN- HOUSED EGGS Cumulative	MORT- ALITY Cumulative (%)	BODY WEIGHT (kg)	BODY WEIGHT (lb)	FEED INTAKE (lb/day/100)	WATER CONS. ¹ (gal/day/100)	HEN- HOUSED EGG MASS Cumulative (lb)	AVG. EGG WEIGHT ² (lb/case)
49	90 – 92	188.5 – 194.5	186.5 – 192.5	2.0	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	24.50	48.7
50	89 – 91	194.7 – 200.9	192.6 – 198.8	2.1	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	25.34	48.8
51	89 – 91	200.9 – 207.3	198.7 – 205.0	2.2	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	26.17	48.8
52	89 – 91	207.2 – 213.7	204.8 – 211.2	2.2	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	27.01	48.9
53	88 – 91	213.4 – 220.0	210.8 – 217.4	2.3	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	27.84	49.0
54	88 – 90	219.5 – 226.3	216.8 – 223.6	2.4	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	28.67	49.0
55	88 – 90	225.6 – 232.6	222.8 – 229.8	2.5	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	29.50	49.1
56	87 – 90	231.8 – 238.9	228.8 – 235.9	2.6	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	30.32	49.1
57	87 – 89	237.8 – 245.1	234.7 – 241.9	2.7	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	31.14	49.2
58	87 – 89	243.9 – 251.3	240.6 – 248.0	2.8	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	31.96	49.2
59	86 – 88	249.9 – 257.5	246.5 – 254.0	2.8	1.51–1.56	3.32– 3.45	21.8–23.1	3.84 – 5.44	32.77	49.3
60	86 – 88	256.0 – 263.7	252.3 – 259.9	2.9	1.51–1.56	3.32– 3.45	21.8–23.1	3.80 – 5.39	33.58	49.3
61	86 – 88	261.9 – 269.8	258.1 – 265.9	3.0	1.51–1.56	3.32– 3.45	21.8–23.1	3.80 – 5.39	34.39	49.4
62	85 – 88	267.9 – 276.0	263.9 – 271.9	3.1	1.51–1.56	3.32– 3.45	21.8–23.1	3.80 – 5.39	35.19	49.4
63	85 – 87	273.9 – 282.1	269.6 – 277.8	3.2	1.51–1.56	3.32– 3.45	21.8–23.1	3.76 – 5.34	36.00	49.5
64	85 – 87	279.8 – 288.1	275.4 – 283.6	3.3	1.51–1.56	3.32– 3.45	21.8–23.1	3.76 – 5.34	36.79	49.5
65	84 – 86	285.7 – 294.2	281.1 – 289.5	3.4	1.51–1.56	3.32– 3.45	21.8–23.1	3.76 – 5.34	37.59	49.6
66	84 – 86	291.5 – 300.2	286.7 – 295.3	3.5	1.51–1.56	3.32– 3.45	21.8–23.1	3.76 – 5.34	38.38	49.6
67	83 – 85	297.3 – 306.2	292.3 – 301.0	3.6	1.51–1.56	3.32– 3.45	21.8–23.1	3.72 – 5.28	39.16	49.7
68	83 – 85	303.1 – 312.1	297.9 – 306.8	3.7	1.51–1.56	3.32– 3.45	21.8–23.1	3.72 – 5.28	39.94	49.7
69	82 – 85	308.9 – 318.1	303.4 – 312.5	3.8	1.51–1.56	3.32– 3.45	21.8–23.1	3.72 – 5.28	40.72	49.8
70	82 – 84	314.6 – 324.0	308.9 – 318.1	3.9	1.51–1.56	3.32– 3.45	21.8–23.1	3.69 – 5.23	41.49	49.8
71	81 – 84	320.3 – 329.8	314.4 – 323.8	4.0	1.51–1.56	3.32– 3.45	21.8–23.1	3.69 – 5.23	42.26	49.9
72	81 – 83	325.9 – 335.7	319.8 – 329.4	4.0	1.51–1.56	3.32– 3.45	21.8–23.1	3.69 – 5.23	43.03	49.9
73	80 – 83	331.6 – 341.5	325.2 – 335.0	4.1	1.51–1.56	3.32– 3.45	21.8–23.1	3.69 – 5.23	43.79	50.0
74	80 – 83	337.2 – 347.3	330.6 – 340.5	4.2	1.51–1.56	3.32– 3.45	21.8–23.1	3.65 – 5.18	44.55	50.0
75	80 – 82	342.7 – 353.0	335.9 – 346.0	4.3	1.51–1.56	3.32– 3.45	21.8–23.1	3.65 – 5.18	45.30	50.0
76	79 – 82	348.3 – 358.8	341.2 – 351.5	4.3	1.51–1.56	3.32– 3.45	21.8–23.1	3.65 – 5.18	46.05	50.0
77	79 – 81	353.8 – 364.5	346.5 – 357.0	4.4	1.51–1.56	3.32– 3.45	21.8–23.1	3.61 – 5.12	46.80	50.0
78	78 – 81	359.3 – 370.2	351.7 – 362.4	4.6	1.51–1.56	3.32– 3.45	21.8–23.1	3.61 – 5.12	47.53	50.0
79	78 – 81	364.7 – 375.8	356.9 – 367.8	4.6	1.51–1.56	3.32– 3.45	21.8–23.1	3.61 – 5.12	48.27	50.0
80	77 – 80	370.1 – 381.4	362.0 – 373.1	4.7	1.51–1.56	3.32– 3.45	21.8–23.1	3.61 – 5.12	49.00	50.0
81	77 – 80	375.5 – 387.0	367.1 – 378.4	4.9	1.51–1.56	3.32– 3.45	21.8–23.1	3.61 – 5.12	49.72	50.0
82	76 – 79	380.8 – 392.6	372.2 – 383.7	5.0	1.51–1.56	3.32– 3.45	21.8–23.1	3.57 – 5.07	50.44	50.0
83	76 – 79	386.1 – 398.1	377.2 – 388.9	5.2	1.51–1.56	3.32– 3.45	21.8–23.1	3.57 – 5.07	51.15	50.0
84	75 – 78	391.4 – 403.6	382.2 – 394.1	5.3	1.51–1.56	3.32– 3.45	21.8–23.1	3.57 – 5.07	51.86	50.0
85	75 – 78	396.6 – 409.0	387.1 – 399.3	5.5	1.51–1.56	3.32– 3.45	21.8–23.1	3.57 – 5.07	52.56	50.0
86	74 – 77	401.8 – 414.4	392.0 – 404.4	5.6	1.51–1.56	3.32– 3.45	21.8–23.1	3.57 – 5.07	53.25	50.0
87	74 – 77	406.9 – 419.8	396.9 – 409.4	5.8	1.51–1.56	3.32– 3.45	21.8–23.1	3.53 – 5.02	53.94	50.1
88	73 – 76	412.1 – 425.1	401.7 – 414.5	6.0	1.51–1.56	3.32– 3.45	21.8–23.1	3.53 – 5.02	54.63	50.1
89	73 – 76	417.1 – 430.4	406.5 – 419.5	6.1	1.51–1.56	3.32– 3.45	21.8–23.1	3.53 – 5.02	55.31	50.1
90	72 – 75	422.2 – 435.7	411.2 – 424.4	6.3	1.51–1.56	3.32– 3.45	21.8–23.1	3.53 – 5.02	55.98	50.1
91	72 – 75	427.2 – 440.9	415.9 – 429.3	6.5	1.51–1.56	3.32– 3.45	21.8–23.1	3.53 – 5.02	56.65	50.1
92	71 – 74	432.2 – 446.1	420.6 – 434.1	6.7	1.51–1.56	3.32– 3.45	21.8–23.1	3.49 – 4.97	57.31	50.1
93	71 – 74	437.1 – 451.3	425.2 – 439.0	6.8	1.51–1.56	3.32– 3.45	21.8–23.1	3.49 – 4.97	57.96	50.1
94	70 – 73	442.0 – 456.4	429.7 – 443.7	7.0	1.51–1.56	3.32– 3.45	21.8–23.1	3.49 – 4.97	58.61	50.1
95	70 – 73	446.9 – 461.5	434.3 – 448.5	7.2	1.51–1.56	3.32– 3.45	21.8–23.1	3.49 – 4.97	59.26	50.1
96	69 – 72	451.8 – 466.6	438.7 – 453.1	7.4	1.51–1.56	3.32– 3.45	21.8–23.1	3.49 – 4.97	59.90	50.1
97	69 – 72	456.6 – 471.6	443.2 – 457.8	7.6	1.51–1.56	3.32– 3.45	21.8–23.1	3.45 – 4.91	60.53	50.1
98	68 – 71	461.3 – 476.6	447.6 – 462.4	7.8	1.51–1.56	3.32– 3.45	21.8–23.1	3.45 – 4.91	61.15	50.1
99	68 – 71	466.1 – 481.6	451.9 – 466.9	7.9	1.51–1.56	3.32– 3.45	21.8–23.1	3.45 – 4.91	61.77	50.1
100	67 – 70	470.8 – 486.5	456.3 – 471.5	8.0	1.51–1.56	3.32– 3.45	21.8–23.1	3.45 – 4.91	62.39	50.1

¹ The chart shows an expected range of feed and water consumption at normal environmental temperatures of 70–81°F. As the environmental temperature increases above this range, water consumption may increase up to double the amounts shown.

² Egg weights after 40 weeks of age assume phase feeding of protein to limit egg size.

Cage Brooding Recommendations

Chick Placement

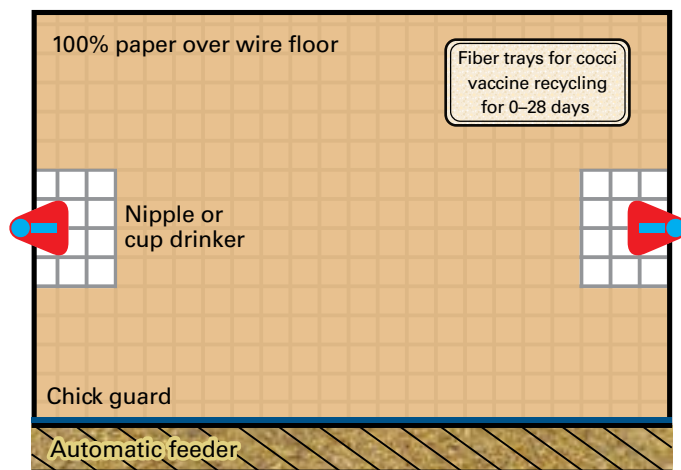
- Unload boxes quickly and gently place chicks in brooding area.
- Brood chicks in groups from similar aged breeder flocks.
- Start chicks in upper tiered cages which are usually warmer and brighter. Ensure there are no shadows on drinkers (3–5 footcandles at the level of the nipple).
- Chicks from young breeder flocks should be placed in warmer and brighter areas of the house.

House Preparation before Arrival of Chicks

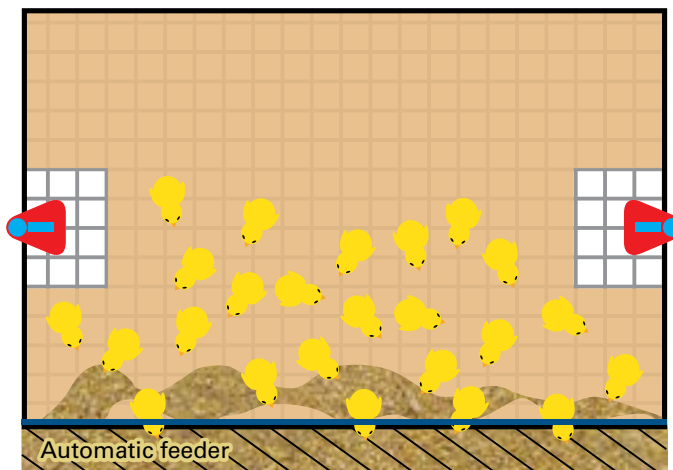
- Brooder house should be completely cleaned and disinfected well in advance of chick delivery. Confirm effectiveness of cleaning and disinfection with environmental swabs.
- Allow 2 weeks minimum downtime between flocks.
- For more information on house preparation and brooding management, see the "[Growing Management of Commercial Pullets](http://www.hylinena.com)" technical update at www.hylinena.com.
- Establish proper house temperature of 92–96°F (air temperature measured at chick level) and 60% humidity 24 hours before chick placement.
- Pre-heat brooding houses prior to chick placement: 24 hours in normal climates, 48 hours in cool climates and 72 hours in cold climates.
- Fill automatic feed line to its highest level and adjust chick guards. Allow access to the automatic feeder from the first day.
- Bright light (3–5 footcandles) for up to 21 days helps chicks quickly find feed and water and adapt to the new environment.

Brooding Management

- Place feed on cage paper 3+ days to encourage consumption. For beak-treated chicks, feed on paper for 7+ days.
- Place feed in front of permanent feeder to train chicks to move toward feeders.
- Remove paper at 7–14 days of age to avoid build-up of feces. If using coccidiosis vaccine, talk to the manufacturer regarding paper management.
- Find optimum balance of temperature, humidity and ventilation rate for chick comfort.
- Cage floors should not be slippery or sloped.
- Use vitamins and electrolytes in chicks' water (avoid sugar-based products to prevent growth of microorganisms).
- Chicks' body weight should double between arrival and 7 days of age.
- Brood chicks in groups from similar aged breeder flocks.
- Modify temperature as needed to meet chicks' comfort needs.
- Adjust brooding temperature according to relative humidity. Lower temperature should be used with higher relative humidity.
- For every 5 percentage point increase above 60% relative humidity, reduce brooding temperatures by 2°F.
- After the first week, reduce temperature weekly 4–6°F until reaching 70°F.
- Do not give cold water to chicks. Be careful when flushing water lines for chicks. Allow water time to warm up in the house so chicks are comfortable drinking.



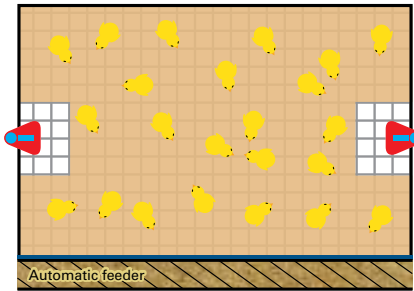
Chick guard adjusted to allow access to feeder from first day



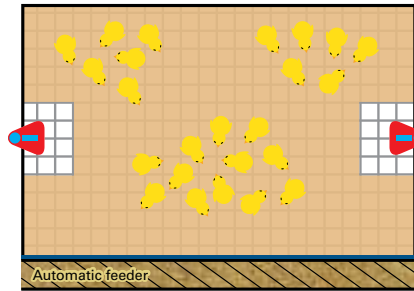
Place feed on paper near automatic feeder to train chicks

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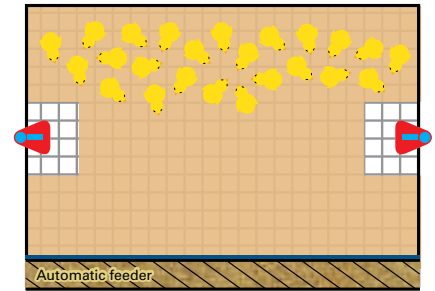
Cage Brooding Recommendations *(continued)*

**CORRECT**

Chicks evenly distributed in cage,
active and sounding content

**COLD**

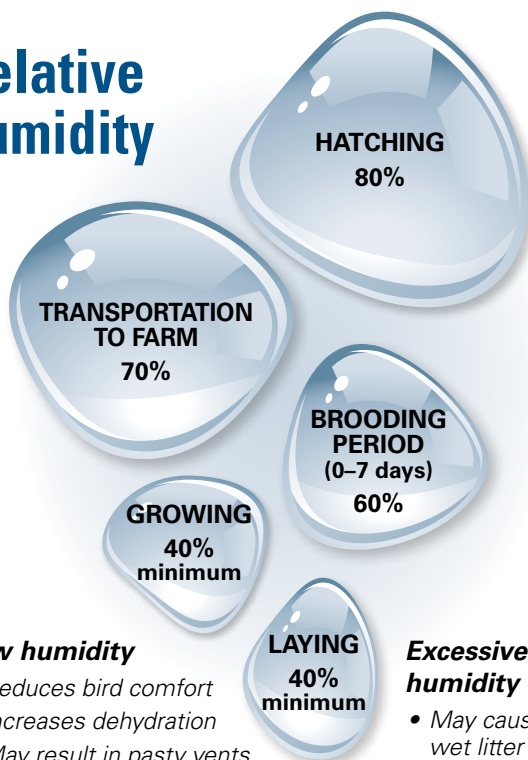
Chicks gathered into groups
sounding distressed

**UNEVEN VENTILATION**

Chicks congregated in one part
of cage, avoiding drafts, noise or
uneven light distribution

AGE	0–3 days	4–7 days	8–14 days	15–21 days	22–28 days	29–35 days	36–42 days
AIR TEMP. (CAGE)	90–92°F						
AIR TEMP. (FLOOR)	91–95°F	86–90°F	82–86°F				
LIGHT INTENSITY	3–5 footcandles	88–91°F	82–86°F				
LIGHT HOURS	22 hours or Intermittent Program	3–5 footcandles 21 hours or Intermittent Program	84–88°F 3–5 footcandles 20 hours	79–82°F 81–84°F 3–5 footcandles 19 hours	73–79°F 75–81°F 1–3 footcandles 18 hours	70–73°F 72–75°F 0.5–1.5 footcandles 17 hours	70°F 70°F 0.5–1.5 footcandles 16 hours

Relative humidity

**Low humidity**

- Reduces bird comfort
- Increases dehydration
- May result in pasty vents in chicks
- May increase agitation and possibility of pecking
- Adversely affects feather cover
- Increases dust

Excessive humidity

- May cause wet litter
- Increases ammonia
- Causes poor air quality

CROP FILL – ARE THE CHICKS EATING?

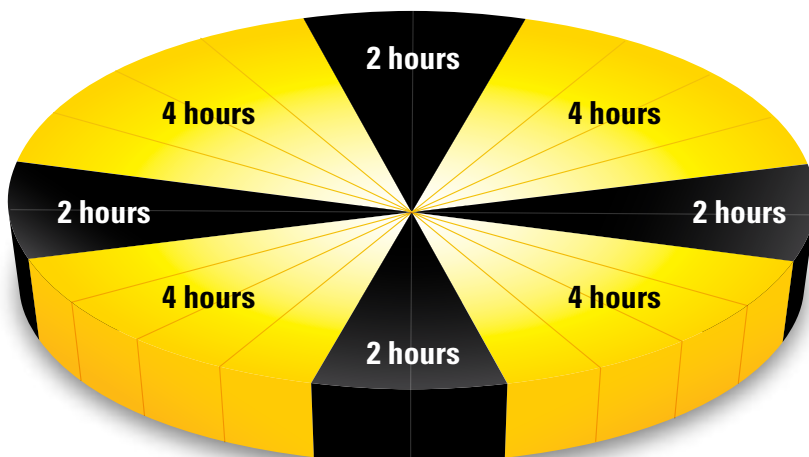
Hours after chick placement	Chicks with feed in crop		
6	75%		
12	85%	Chick with starter feed in crop	Chick without starter feed in crop
24	100%		

Brooding temperatures that are too low or too high will decrease the percentage of chicks with crop fill.

Lighting During Brooding Period

- An intermittent lighting program is preferred. If not using an intermittent lighting program from 0–7 days, then use 20 hours of light from 0–7 days.
- Do not use 24 hours of light.
- Bright light (3–5 footcandles) for up to 21 days helps chicks quickly find feed and water and adapt to the new environment.
- If multiple ages of chicks are moved into the house, use the intermittent lighting program based on youngest chicks. Oldest chicks can have up to 14 days of intermittent lighting.
- After 2–3 weeks, begin slow step-down lighting program (see [Light Program for Light-Controlled Housing](#)).

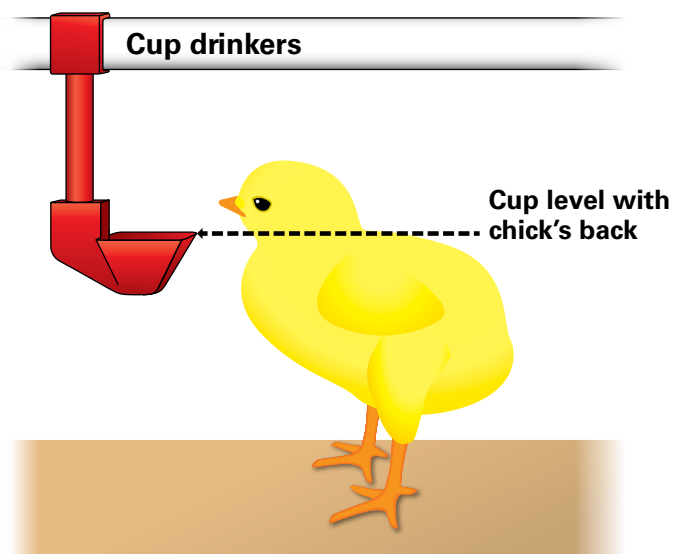
INTERMITTENT LIGHTING PROGRAM



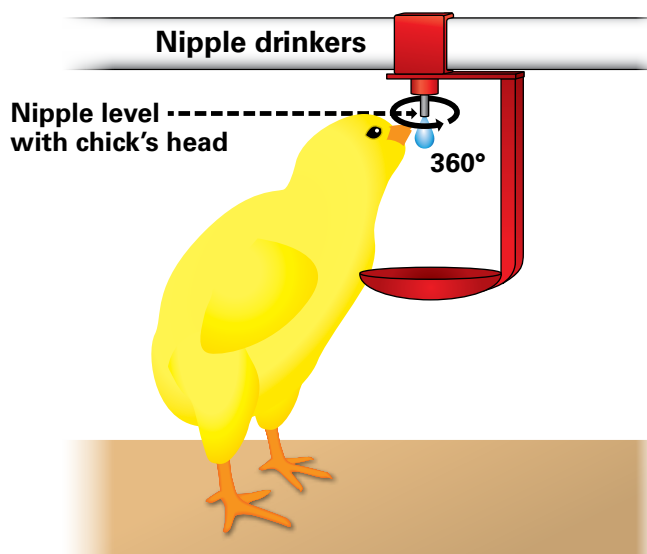
- Preferred lighting technique
- Use from 0–7 days
- Intermittent dark periods provide rest periods for chicks
- Synchronizes chicks' activities and feedings
- Establishes more natural behavior of rest and activity
- May improve 7 day livability and pullet body weight
- Some dark periods may be shortened or removed to accommodate work schedules

Drinking Systems

- The type of drinkers used during rearing should be the same as in the layer house. 360 degree nipples are strongly preferred. Nipple type in rearing and laying house (vertical vs. 360° nipples) should be the same.
- Drinking water should be tested periodically for quality and cleanliness from source and end of the water line.
- Flush water lines prior to chick arrival.
- Flush water lines weekly during rearing and production periods.
- Record daily flock water consumption. A drop in water consumption is often the first sign of a serious problem in the flock.



- Cup drinkers should be manually filled during 0–3 days to train chicks to drink.
- Open drinkers (bell, plasjon, supplemental chick drinkers, trough) are easily contaminated and should be cleaned daily.



- Adjust nipple water system pressure to create hanging drop to help chicks find water for 0–3 days and in layer house at transfer.
- Splash cups are useful during brooding period and in hot climates.
- Use only 360° activated nipples for hatchery beak-treated chicks.

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Beak Treatment / Trimming

(Check local regulations concerning use of beak trimming)

- Hy-Line W-36 commercial layers can be successful without beak trimming in cage and cage-free environments where light intensities are well controlled.
- Infrared beak treatment (IRBT) is the preferred method of beak modification for Hy-Line W-36 commercial layers.
- If IRBT is not used, 7–10-day precision beak trimming is acceptable.
- If necessary, re-trim at 6 weeks or 12–14 weeks of age.

HATCHERY INFRARED BEAK TREATMENT (IRBT)

- This provides efficient, uniform beak treatment.
- Beak remains intact until 21–28 days of age when treated portion separates.
- Use only 360° activated nipples for IRBT chicks, as well as supplemental chick drinkers.
- For IRBT chicks, feed on paper for 7–10 days.
- In open housing, IRBT treated pullets may need retrimming at 6 or 12–14 weeks.
- For more information, see the “[Infrared Beak Treatment](#)” technical update at www.hylinena.com.



Infrared beak treatment can be modified according to local conditions.



Immediately following infrared beak treatment on day of hatch



7 days after infrared beak treatment

Precautions when using IRBT or beak trimming birds:

- Water intake and light intensity are the most important factors in the success of IRBT/beak trimming. Chicks require immediate and easy access to water.
- Do not beak-trim sick or stressed birds.
- Do not hurry; handle chicks carefully.
- Provide vitamins and electrolytes containing vitamin K in drinking water 2 days before and 2 days after beak trimming.
- Watch chicks after beak trimming to assess stress. Raise ambient temperature until birds appear comfortable and active.
- Keep feed at the highest level for several days after beak trimming.
- Use only well-trained crews.
- Use 360° activated nipples, supplemental chick drinkers and splash cups to encourage drinking.



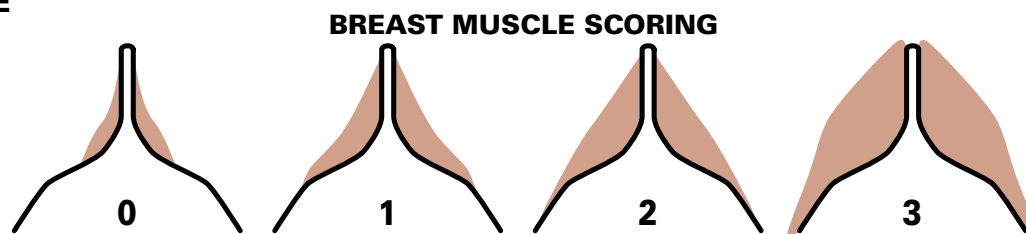
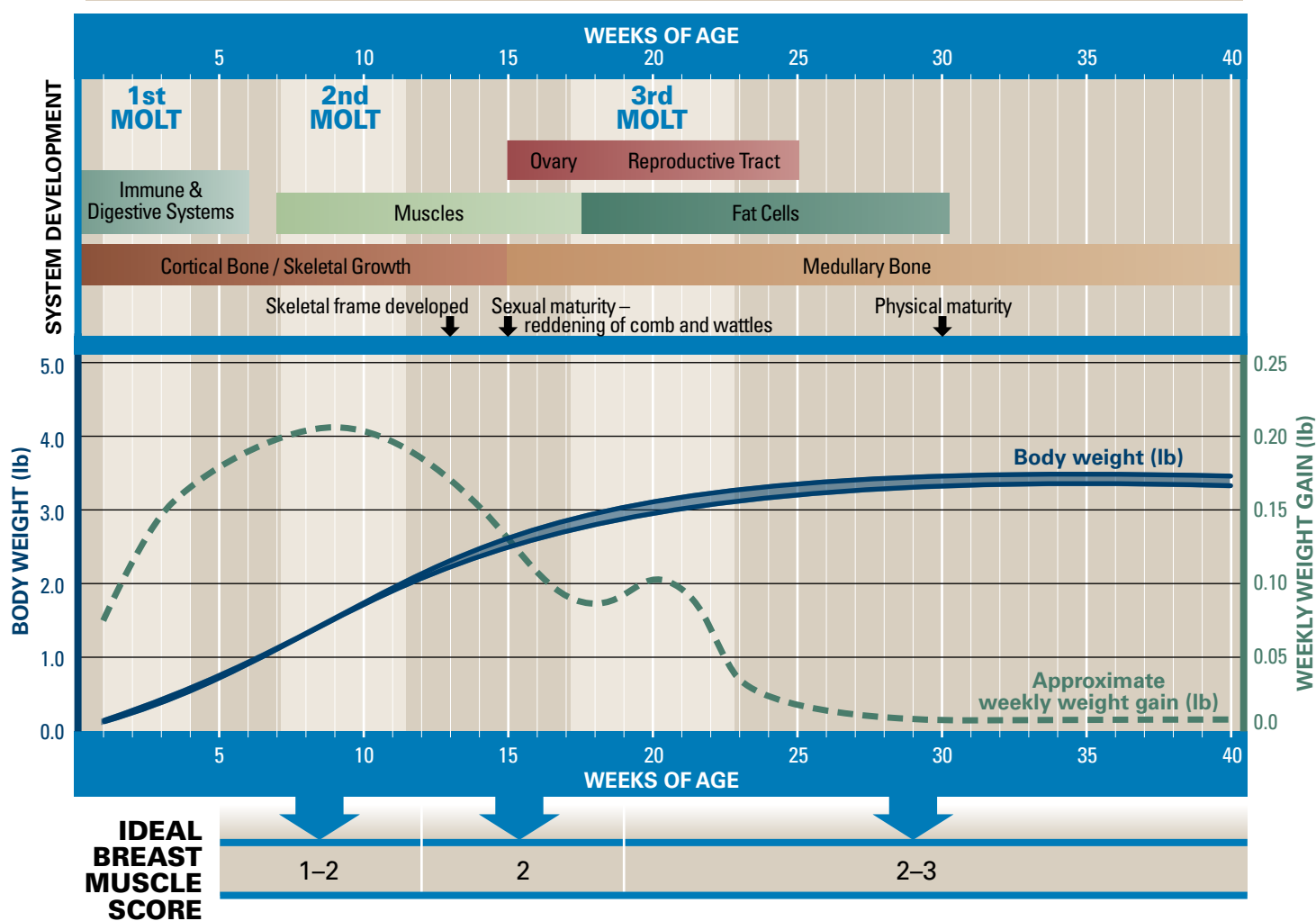
IRBT



Beak trim

Growth and Development

- Focus on pullet rearing programs to optimize growth and development.
- The best predictor of future laying performance is the pullets' body weight and body type at the point of lay.
- A pullet flock stimulated into egg production at the correct body weight (2.75–2.85 lb / 1247–1293 g) with uniformity higher than 90% performs best in the production period.
- It is important to achieve 6, 12, 18, 24, and 30 week body weight targets to ensure optimum development of the bird's body.
- Manage feeding to achieve body weights 0–12 weeks in the upper range of the body weight standard. This will ensure good musculoskeletal and GIT development. Avoid high rates of body weight gains after 12 weeks to prevent excessive abdominal fat development.
- Delay diet change if birds are underweight or have poor diet uniformity.
- Anticipate rapid rise in ambient temperature and adjust bird's diet accordingly. Birds will eat less when exposed to a rapid temperature increase. (See the "[Understanding Heat Stress in Layers](#)" technical update at www.hylinena.com.)
- Stress periods require a change in diet formulation to ensure proper nutrient intake.
- Increasing dietary fiber to 5–6% beginning with developer diet (after the flock has reached 2.09–2.14 lb / 950–970 g) can increase crop, gizzard and intestine development and capacity.



Layers with good muscle development are better able to sustain high egg production

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Growing Body Weights, Feed Consumption and Uniformity

Weigh 100 birds weekly to 30 weeks of age

AGE (weeks)	BODY WEIGHT (g)	BODY WEIGHT (lb)	FEED INTAKE (lb/day per 100 birds)	WATER CONSUMPTION (gal/day/100)	UNIFORMITY (Cage)
1	61 – 69	0.13 – 0.15	2.89 – 3.50	0.52 – 0.84	>85%
2	112 – 118	0.25 – 0.26	3.68 – 4.48	0.66 – 1.07	
3	176 – 184	0.39 – 0.41	4.78 – 5.83	0.86 – 1.40	
4	245 – 255	0.54 – 0.56	6.29 – 7.39	1.13 – 1.77	>80%
5	324 – 336	0.71 – 0.74	7.50 – 8.40	1.35 – 2.01	
6	413 – 427	0.91 – 0.94	8.70 – 9.53	1.56 – 2.28	
7	502 – 518	1.11 – 1.14	9.40 – 10.30	1.69 – 2.47	
8	591 – 609	1.30 – 1.34	10.13 – 11.26	1.82 – 2.70	
9	680 – 700	1.50 – 1.54	11.11 – 12.08	2.00 – 2.90	
10	779 – 801	1.72 – 1.77	11.82 – 12.81	2.12 – 3.07	
11	868 – 892	1.91 – 1.97	12.19 – 13.15	2.19 – 3.15	
12	947 – 973	2.09 – 2.15	12.43 – 13.51	2.23 – 3.24	>85%
13	1016 – 1044	2.24 – 2.30	12.86 – 14.15	2.31 – 3.39	
14	1085 – 1115	2.39 – 2.46	13.06 – 14.56	2.35 – 3.49	
15	1154 – 1186	2.54 – 2.61	13.41 – 14.90	2.41 – 3.57	>90%
16	1193 – 1227	2.63 – 2.71	14.00 – 15.20	2.52 – 3.64	
17	1232 – 1268	2.72 – 2.80	14.75 – 15.90	2.65 – 3.81	



Weigh birds individually from 0–30 weeks using a digital scale that calculates uniformity.

- Monitor body weights weekly from 0–30 weeks and before scheduled diet changes.
- Body weight gains and uniformity may be negatively affected by inappropriate diet changes, bird handling, vaccination and transfer.
- If using multiple hatch dates in the same house (which creates an age range), weigh birds from the same hatch date.
- Flocks should be at 90% uniformity at the time of transfer to the laying facility.
- During the transfer of birds from rearing to laying facilities, there will be some loss of body weight.

Space Guidelines

At the time of this writing, the most recent guidelines can be found at <https://uepcertified.com/uep-certified-resources/>.

Transfer to Laying House

- The flock can be moved into the production facility at 15–16 weeks of age.
- It is ideal to administer the last inactivated vaccines at least one week prior to transfer.
- Delayed transfer results in overcrowding in rearing cages and loss of flock uniformity and body condition.
- It is important that rearing and production cages use similar feed and water systems, to minimize stress.
- Any sex slips (males) should be removed around 7 weeks and at transfer.
- Supportive care to reduce stress, such as water-soluble vitamins, probiotics, vitamin C and increased dietary density, should be used 3 days before and 3 days after transfer.
- Weigh prior to transfer and monitor weight loss during transfer. Birds can lose up to 0.25 lb (113 g) of body weight during transfer, but should recover this loss within one week.
- Monitor flock water consumption frequently after transfer. Pre-transfer water consumption should be achieved within 6 hours after transfer to the laying house.
- Brighten the lights for three days after transfer until birds settle in their new environment.
- Inspect the flock and remove mortality daily.
- If mortality exceeds 0.1% per week, perform necropsies and other diagnostics to determine cause(s) of mortality.
- Transfer birds as quickly as possible to laying house. Transfer all birds the same day. Move early in the morning so birds can keep to a normal daily routine.

Enriched Cages

- Enriched cages address some of the welfare concerns of layers in cages by providing more space with environment enrichment devices, such as perches, nest boxes, scratch areas and abrasive pads for beak and toe management.
- Generally, bird group sizes range from 40–110 birds per cage.
- As group size increases, there is more competition for feed and water space and less stable social groups. This could lead to behavioral problems like feather pecking and piling. Cage enrichments help prevent these behavioral problems.

Enrichable Cages

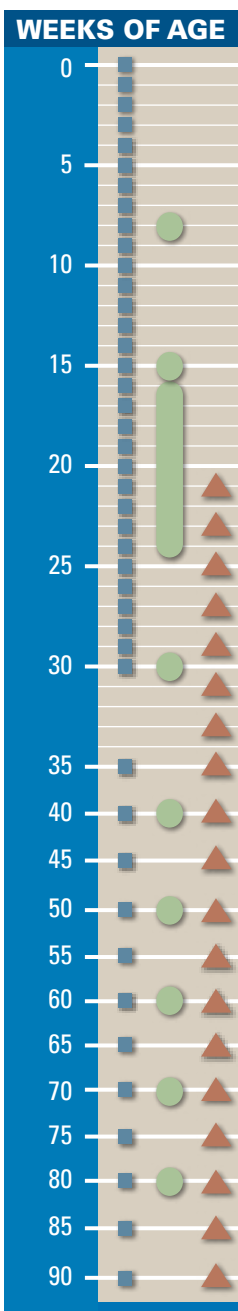
- Enrichable cages are colony cages providing more space without any of the environmental enrichment devices, but are designed so that the environmental enrichments can be added at a later date.
- Bird group sizes range from 40–110 birds per cage.
- As group size increases, there is more competition and less stable social groups. This could lead to behavioral problems like feather pecking and piling. Cage enrichments, especially perches, help prevent these behavioral problems.
- Without cage enrichment devices, keep bird group size in enrichable cages to less than 40 birds per cage.

Alternative Systems

- W-36 is an excellent choice for alternative management systems, including aviaries. For more information on managing the W-36 in such systems, see the [Alternative Systems Management Guide](#) at www.hylinena.com.

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Management Events for Commercial Layers



AGES OF BODY WEIGHT MEASUREMENTS

- Weigh separate groups of birds on each cage level due to temperature and environmental differences. If using enrichable or enriched cages, weigh all birds in the cage, in at least three locations (may be more than 100 birds).
- Identify cages from the beginning and end of feed lines.
- Mark cages and use the same cage every time body weight is monitored.
- Weigh birds on the same day of the week and at the same time of day.

0–30 weeks

- Weigh 100 birds individually every week.
- Weigh birds in the same cages each time for best accuracy.
- Calculate uniformity.

35–90 weeks

- Weigh 100 birds individually every 5 weeks.
- Weigh birds in the same cages each time for best accuracy.
- Calculate uniformity.

When handling birds for body weights, assess:

- Keel bone—straightness and firmness
- Breast muscle score
- Body fat
- External parasites
- Clinical symptoms of disease

CALCULATING UNIFORMITY

- Use individual bird weights.
- Uniformity calculation tool is available at www.hylinena.com.

AGES OF SERA COLLECTION

For more information, see the “[Proper Collection and Handling of Diagnostic Samples](#)” technical update at www.hylinena.com.

Collect 10–20 sera samples per flock for titer determination.

8 weeks

- Assess early vaccination technique and disease exposure.

15 weeks

- Collect sera before transfer to lay house to assess possible change in disease exposure.
- It is common to not send sera to the laboratory and freeze for future analysis in event of disease outbreak on lay farm.

16–24 weeks

- Collect sera at least 4 weeks after final inactivated vaccination to measure post-vaccination antibody response.
- It is useful to assess disease challenge after transfer to lay farm.

30–80 weeks

- Collect sera every 10 weeks.
- It is useful for assessing disease exposure during the laying period.

AGES TO MONITOR EGG WEIGHTS

Weigh 100 eggs collected from egg belts in front of randomly selected cages (may be the same cages used for body weight monitoring) to ensure even distribution of egg samples. Monitor egg weights on a specific day of the week within the same 3-hour time frame.

Bird Handling—Be Gentle

- Proper handling of birds during body weight measurements, blood collection, selection, vaccination, and transfer will reduce bird stress and prevent injuries.
- Hold birds by both legs or both wings.
- Return birds to cage or floor gently.
- Use experienced personnel that have been trained in proper procedures of bird handling.
- Continually observe crews for proper handling.

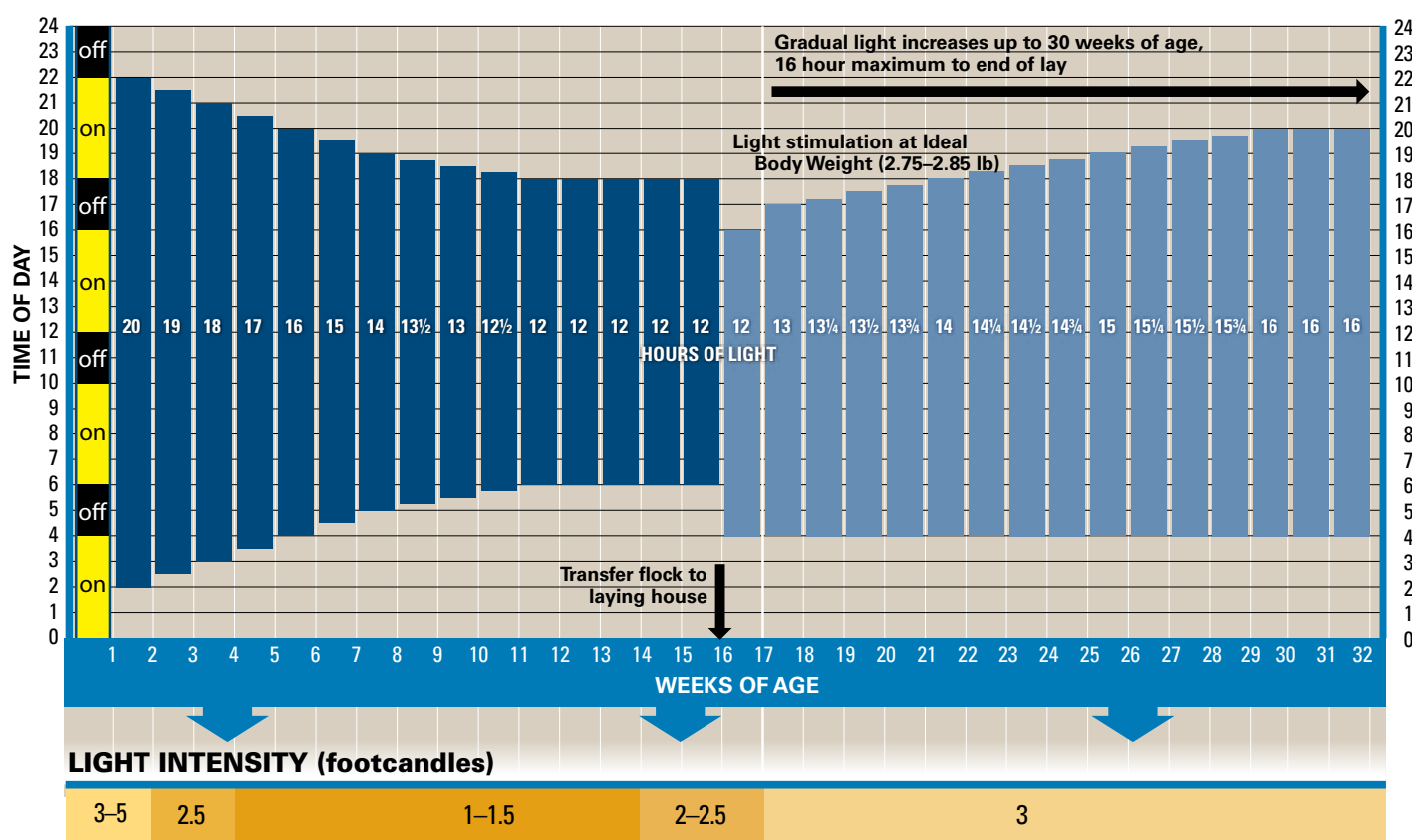


Hold no more than three birds in one hand.

Good Lighting Practices

- Measure minimum light intensity at feeder on bottom tier cages, mid-way between lights.
- Keep light bulbs clean to prevent loss of light intensity.
- Prevent dark areas caused by too much distance between lights or burned-out light bulbs.
- Position lights to minimize bright and dark areas in the house.
- Shiny or white surfaces reflect light and increase light intensity.
- Take local conditions into account which may require adaptations of lighting programs.
- Light hours of rearing and production houses should be matched at transfer.
- Light intensity should gradually increase 2 weeks before flock is transferred to the laying house (but not prior to 14 weeks of age). Final rearing house light intensity should match the laying house intensity.
- Begin light stimulation when flock reaches the body weight target of 2.75–2.85 lb (1247–1293 g). Delay light stimulation if the flock is underweight or has poor uniformity.
- Light stimulation period should extend into the peaking period. Achieve 16 hours of light at approximately 30 weeks.
- Alternating the height of lights improves light distribution to all cage levels

Light Program for Light-Controlled Housing



- Light-controlled houses are those which use light traps around fans and air inlets and complete prevent the ingress of light from the outside. Houses that are not light controlled should use lighting programs for open-sided housing.
- An intermittent lighting program for chicks is preferred. If not using an intermittent lighting program from 0–7 days, then use 20 hours of light from 0–7 days.
- “Lights on” time can be varied between houses in laying flocks to facilitate egg collection on multiple flock complexes.
- If the laying flock has a large spread in hatch ages and/or poor uniformity, light stimulate the flock based on the youngest hatch date or lightest birds.
- Up to 16 weeks, use cool light (4000–6000 K).
- During lay, use warm lights (2700–3500 K) to ensure sufficient red spectrum light.
- For more information on poultry lighting, see the “[Understanding Poultry Lighting](#)” and “[Impact of Tarp Color on Poultry Lighting](#)” technical updates at www.hylinena.com.

The Hy-Line International Lighting Program can create custom lighting programs for your location, hatch date, and variety.

- On the first screen—enter e-mail address and select language.
- On the second screen, use dropdowns for “Select Location of Flock,” “Hatch Date,” “Variety Standards” and “Housing Style.”
- Click on “Create Lighting Spreadsheet.”
- Results will be displayed on the page; you can also download them in Excel.

Hy-Line Lighting Program for ICMA / DALLAS CENTER 32° 56' W 41° 42' N

Hy-Line W-58 Commercial (Dark grows dark by)

Match Date 1/3/2029

T
I
M
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O
F
D
A
Y

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

1/3/2029 1/31/2029 2/28/2029 3/27/2029 4/24/2029 5/22/2029 6/19/2029 7/17/2029

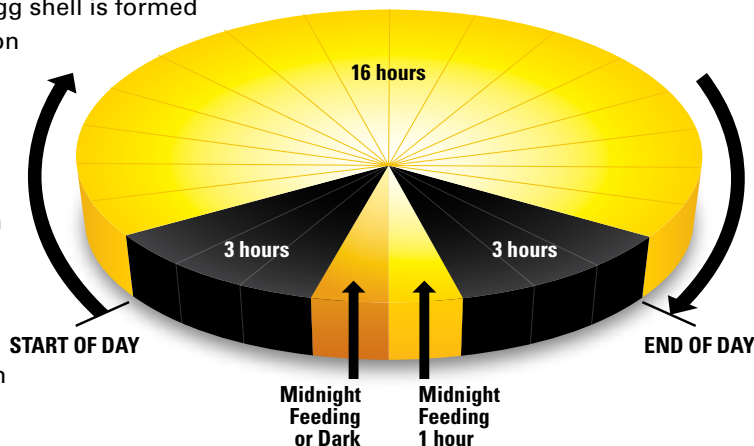
Age in Weeks

Midnight Feeding / Lighting Program

- Optional lighting technique that promotes greater feed consumption
- Used whenever more feed intake is desired in growing or laying flocks
- Increases calcium absorption during night when most egg shell is formed
- Useful to increase feed intake during peak egg production
- Helps maintain feed consumption in hot climates
- Midnight feeding may increase feed intake 0.5–1.0 lb / day per 100 birds

Good Practices

- Initiate the program by turning lights on for 1–2 hours in the middle of the dark period.
- Fill feeders before lights are turned on.
- There must be at least 3 hours of dark before and after the midnight feeding.
- Light provided during the midnight feeding is in addition to regular day length (i.e. 16 hours + midnight feeding).
- If midnight feeding is removed, reduce light gradually at a rate of 15 minutes per week.



Heat Stress

For information on management of layers in heat stress conditions, see the “[Understanding Heat Stress in Layers](#)” technical update at www.hylinena.com.

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Water Quality

- Good quality water must be available to birds at all times.
- Water and feed consumption are directly related—when birds drink less, they consume less feed and production quickly declines.
- As a general rule, healthy birds will consume 1.5–2.0 times more water than feed. This ratio increases in high ambient temperatures.
- Test water quality at least 1 time per year. The water source will determine the regularity of water testing.
 - Surface water requires more frequent testing, as it is more affected by season and rainfall patterns.
 - Closed wells taking water from aquifers or deep artesian basins will be more consistent in water quality, but are generally higher in dissolved mineral content.
- The presence of coliform bacteria is an indicator that the water source has been contaminated with animal or human waste.
- When collecting a well water sample, let the water run for 2 minutes prior to collecting the sample. Water samples should be kept below 50°F and submitted to the lab in less than 24 hours.
- Some water sources contain high levels of dissolved minerals such as calcium, sodium and magnesium. When this occurs, amounts of these minerals in water have to be considered when formulating feed.
- Ideal water pH is 5–7 to promote good water sanitation, increase feed consumption and improve upper gastrointestinal health.
- Less than optimum water quality can have a significant impact on gut health, which will lead to under utilization of nutrients in feed.
- A decrease in flock water consumption is often the first sign of disease problems and production drops.

ITEM	MAXIMUM CONCENTRATION (ppm or mg/L)*	
Nitrate NO_3^- ¹	25	Older birds will tolerate higher levels up to 20 ppm. Stressed or diseased challenged birds may be more sensitive to effects of Nitrate.
Nitrate Nitrogen ($\text{NO}_3\text{-N}$) ¹	6	
Nitrite NO_2^- ¹	4	Nitrite is considerably more toxic than Nitrate, especially for young birds where 1 ppm Nitrite may be considered toxic.
Nitrite Nitrogen ($\text{NO}_2\text{-N}$) ¹	1	
Total dissolved solids ²	1000	Levels up to 3000 ppm may not affect performance but could increase manure moisture.
Chloride (Cl^-) ¹	250	Levels as low as 14 mg may be problematic if sodium is higher than 50 ppm.
Sulfate (SO_4^{2-}) ¹	250	Higher levels may be laxative.
Iron (Fe) ¹	<0.3	Higher levels result in bad odor and taste.
Magnesium (Mg) ¹	125	Higher levels may be laxative. Levels above 50 ppm may be problematic if sulphate levels are high.
Potassium (K) ²	20	Higher levels may be acceptable depending on sodium level, alkalinity and pH.
Sodium (Na) ^{1,2}	50	Higher concentration is acceptable but concentrations above 50 ppm should be avoided if high levels of chloride, sulphate or potassium exist.
Manganese (Mn) ³	0.05	Higher levels may be laxative.
Arsenic (As) ²	0.5	
Fluoride (F^-) ²	2	
Aluminum (Al) ²	5	
Boron (B) ²	5	
Cadmium (Cd) ²	0.02	
Cobalt (Co) ²	1	
Copper (Cu) ¹	0.6	Higher levels result in bitter taste.
Lead (Pb) ¹	0.02	Higher levels are toxic.
Mercury (Hg) ²	0.003	Higher levels are toxic.
Zinc (Zn) ¹	1.5	Higher levels are toxic.
pH ¹	6.3–7.5	Birds may adapt to lower pH. Below pH 5 may reduce water intake and corrode metal fittings. Above pH 8 may reduce intake and reduce effectiveness of water sanitation.
Total bacteria counts ³	1000 CFU/ml	Likely to indicate dirty water.
Total Coliform bacteria ³	50 CFU/ml	
Fecal Coliform bacteria ³	0 CFU/ml	
Oxygen Reduction Potential (ORP) ³	650–750 mEq	The ORP range at which 2–4 ppm of free chlorine will effectively sanitize water at a favorable pH range of 5–7.

*Limits may be lower as interactions exist between magnesium and sulphate; and between sodium, potassium, chloride and sulphate.

¹ Carter & Sneed, 1996. Drinking Water Quality for Poultry, Poultry Science and Technology Guide, North Carolina State University Poultry Extension Service. Guide no. 42

² Marx and Jaikaran, 2007. Water Analysis Interpretation. Agri-Facts, Alberta Ag-Info Centre. Refer to <http://www.agric.gov.ab.ca/app84/rwqit> for online Water Analysis Tool

³ Watkins, 2008. Water: Identifying and Correcting Challenges. Avian Advice 10(3): 10-15 University of Arkansas Cooperative Extension Service, Fayetteville

Air Quality

Air Movement (ft³ / minute per 1000 birds)

AMBIENT TEMPERATURE (°F)	WEEKS OF AGE					
	1	3	6	12	18	19+
90	200	300	600	1500	3500	4500–1500
70	100	150	300	750	1500	2500–3000
50	70	100	200	400	1100	1500–2000
32	40	75	140	300	750	500–750
10	40	60	100	200	300	350–500
-10	40	60	100	200	300	350–400

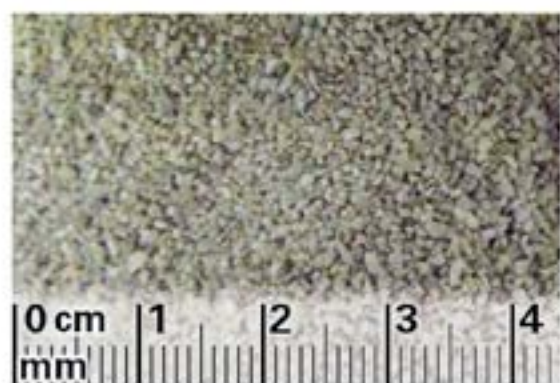
Acknowledgment: Dr. Hongwei Xin, Professor

- Production house should be 64–77°F and 40–60% humidity.
- The general rule for determining required fan capacity—1.0–1.5 ft³ / minute per pound of body weight.
- Ventilation is essential to:
 - Remove moisture from house
 - Remove excessive heat
 - Provide each bird with adequate supply of oxygen
 - Remove carbon dioxide produced by birds
 - Remove dust particles
 - Dilute aerosolized pathogenic organisms
- Allowable levels of gases at floor level in the house are: ammonia (NH₃) < 25 ppm; carbon dioxide (CO₂) < 5000 ppm; carbon monoxide (CO) < 50 ppm.

Calcium Particle Size

PARTICLE SIZE	STARTER, GROWER, DEVELOPER	FIRST EGG-PEAK	PEAK-35 WEEKS	WEEKS 36–55	WEEKS 56+
Fine (0–2 mm)	100%	50%	50%	40%	30%
Coarse (2–4 mm)	–	50%	50%	60%	70%

- The appropriate particle size depends on the solubility of limestone.
- Dietary calcium levels may need to be adjusted based on limestone solubility.
- Limestone dark in color is geologically older, containing more impurities (typically magnesium) and is generally lower in solubility and calcium availability.
- Oyster shell and other marine shells are good sources of soluble calcium.



Fine calcium (0–2 mm)



Coarse calcium (2–4 mm)

Photos courtesy of Longcliff Quarries Ltd.

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Feed Particle Size

A sieve shaker separates feed sample into categories based on particle size.

- Use on the farm to check feed particle size from the feed mill—sample taken on delivery or from feed bins.
- Use to assess the uniformity of feed particle size throughout the feeding system—samples are taken from various points.

Too many fine feed particles:

- Feed intake and nutrient absorption decreases
- Dust in house increases

Too many coarse feed particles:

- Birds selectively eat large particles
- Risk of feed separation increases



Hy-Line Sieve Shaker

OPTIMAL FEED PARTICLE PROFILE

PARTICLE SIZE	STARTER	GROWER	DEVELOPER	PRODUCTION
< 1 mm	1–3 mm diameter, crumble feed should contain < 10% fine feed particles	< 15%	< 15%	< 15%
1–2 mm		45–60%	25–35%	20–30%
2–3 mm		10–25%	25–40%	30–40%
> 3 mm		–	5–10%	10–15%

For more information, see the “[Feed Granulometry](http://www.hylinena.com)” technical update at www.hylinena.com.

Best Practices

- A 3–4 hour gap between mid-day feedings allows birds to consume fine particles. Daily consumption of fine feed particles is important for a balanced nutrient intake.
- Add a minimum of 0.5% liquid oil/fat in meal diets to incorporate and retain small particles in feed.
- Use larger particle size meal or crumble to increase intakes in hot climates.

Vitamins and Trace Minerals

- As the vitamin / trace mineral premix is often found in fine feed particles, a minimum level of 0.5% added liquid oil / fat in meal diets binds small particles in feed.

ITEM ^{1,2,3,4}	IN 2000 LB COMPLETE DIET	
	Growing Period	Laying Period
Vitamin A, IU	9,071,847	7,257,478
Vitamin D ₃ ⁵ , IU	2,993,710	2,993,710
Vitamin E, g	23	18
Vitamin K (menadione), g	3.2	2.3
Thiamin (B ₁), g	2.0	2.3
Riboflavin (B ₂), g	6.0	5.0
Niacin (B ₃) ⁶ , g	36.3	27.2
Pantothenic acid (B ₅), g	9.1	7.3
Pyridoxine (B ₆), g	4.1	3.6
Biotin (B ₇), mg	90.7	68.0
Folic acid (B ₉), g	0.9	0.8
Cobalamine (B ₁₂), mg	20.9	20.9
Manganese ⁷ , g	81.6	81.6
Zinc ⁷ , g	77.1	72.6
Iron ⁷ , g	27.2	36.3
Copper ⁷ , g	13.6	7.3
Iodine, g	1.4	1.1
Selenium ⁷ , g	0.2	0.2

¹ Minimum recommendations for growing and laying periods. Local regulations may limit dietary content of individual vitamins or minerals.

² Store premixes according to supplier's recommendations and observe 'use by' dates to ensure vitamin activity is maintained. Inclusion of antioxidant may improve premix stability.

³ Vitamin and mineral recommendations vary according to activity.

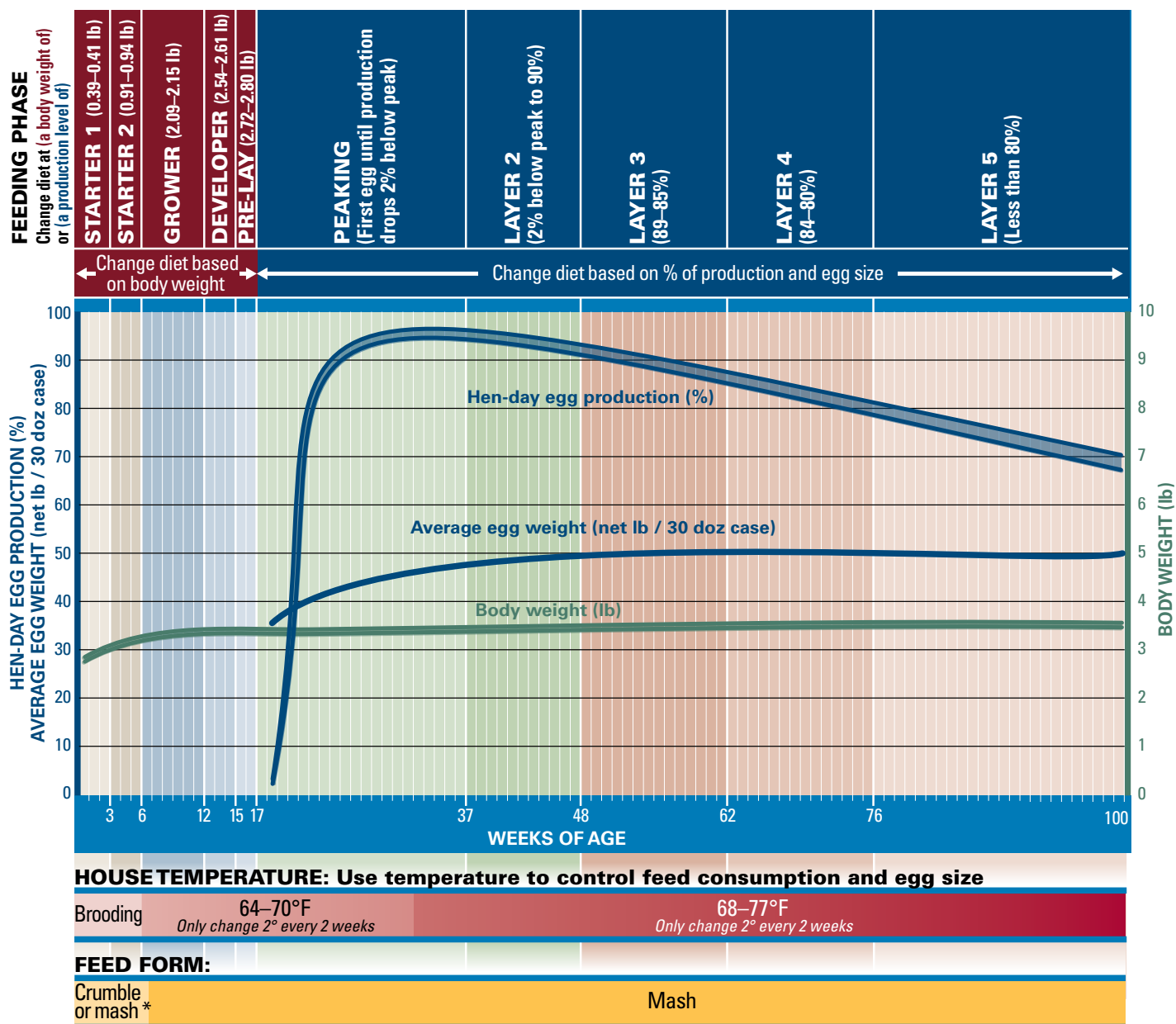
⁴ Where heat treatment is applied to diet, higher levels of vitamins may be required. Consult with vitamin supplier regarding stability through individual production processes.

⁵ A proportion of Vitamin D₃ can be supplemented as 25-hydroxy D₃ according to supplier's recommendations and applicable limits.

⁶ Higher levels of Niacin are recommended in non-cage systems.

⁷ Greater bioavailability and productivity may be possible with use of chelated mineral sources.

Phase Feeding to Meet the Hen's Nutritional Needs



* Crumble may be fed longer to encourage body weight gain

Control of Ambient House Temperature

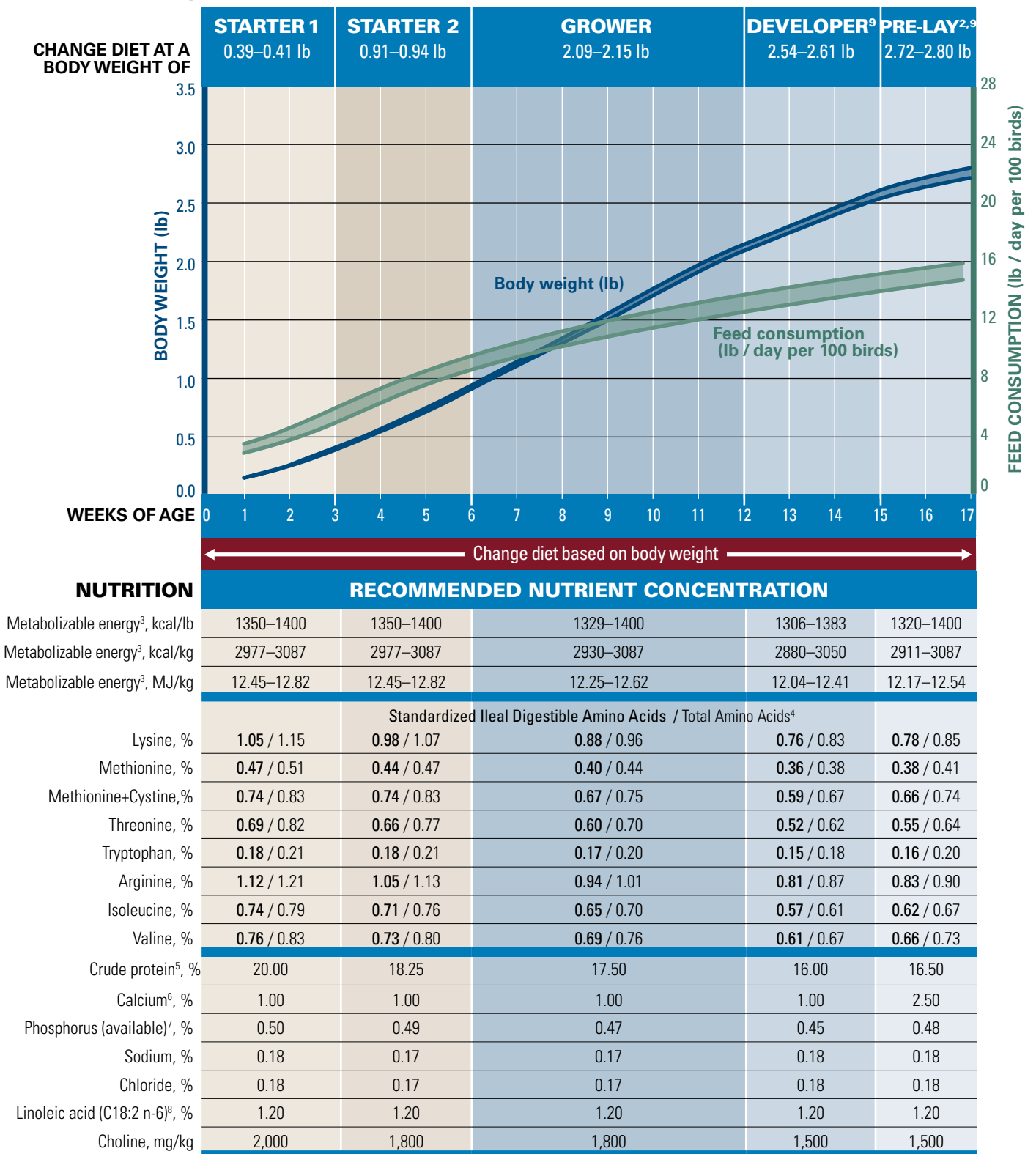
- At housing, an ambient temperature of 64–68°F is desired. Increase house temperature about 2°F every 2 weeks until reaching 77°F assuming ventilation systems are able to maintain adequate air quality at these temperatures.
- Lower (colder) house temperatures after peak will lead to greater feed intakes and may be counterproductive to egg-weight control, as well as optimal feed efficiency and adult hen body weights.
- Place temperature sensors to measure temperature inside of cage. The temperature in walkways is significantly colder than the temperature inside cages, especially in stack deck belted house systems.
- High environmental temperatures have a depressing effect on feed intake.

Controlling Egg Weight

- Closely monitor egg weight of each flock and make nutritional changes as needed to ensure optimal egg weight.
- If smaller eggs are desired, egg weight should be controlled at an early age.
- Egg-weight control is achieved by limiting amino acid consumption and ensuring that feed intake is not too high.
- Monitor egg weight every 2 weeks until 35 weeks of age, then every 5 weeks. Start controlling egg weight when average egg weight is within 1.4 lb / case of target.
- For more information, see the “[Optimizing Egg Size in Layers](#)” technical update at www.hylinena.com.

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Growing Period Nutritional Recommendations¹



¹ All nutrient requirements are based on the feed ingredient table at the back of this guide.

² Do not feed Pre-Lay Diet earlier than 15 weeks of age. Do not feed Pre-Lay later than first egg as it contains insufficient calcium to support egg production. Use Pre-Lay Diet to introduce large particle calcium.

³ Recommended energy range is based on raw material energy values shown in feed ingredient table at back of this guide. It is important that target concentrations of dietary energy are adjusted according to energy system applied to raw material matrix.

⁴ Recommendation for Total Amino Acids is only appropriate to corn and soybean meal diet. Where diets utilize other ingredients, recommendations for Standardized Ileal Digestible Amino Acids must be followed.

⁵ Diets should always be formulated to provide required intake of amino acid. Concentration of crude protein in diet will vary with raw materials used. Crude protein value provided is an estimated typical value only.

⁶ Calcium should be supplied as fine calcium carbonate (mean particle size less than 2 mm). Coarse limestone (2–4 mm) can be introduced in Pre-Lay Diet at up to 50% of total limestone.

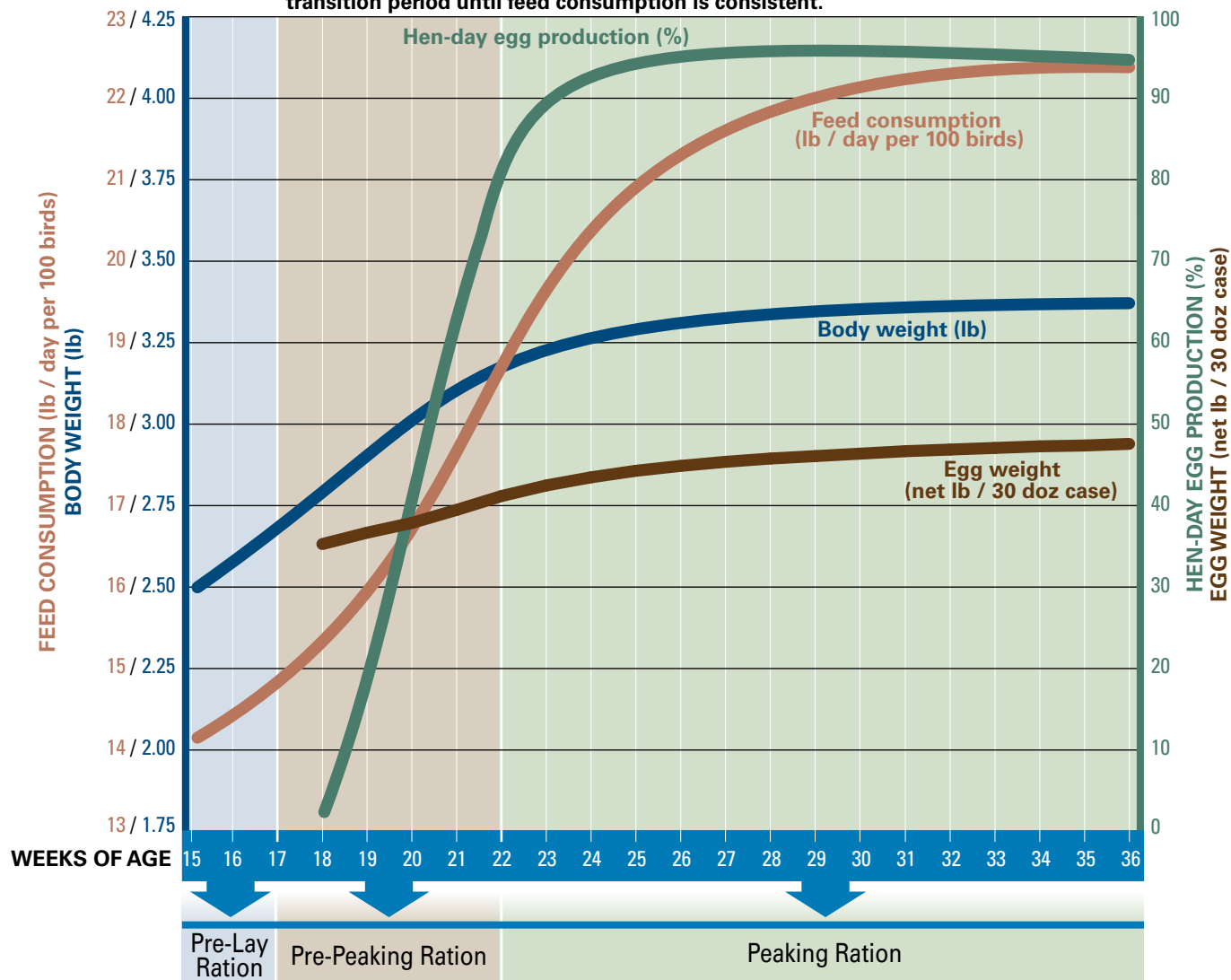
⁷ Where other phosphorus systems are used, diets should contain recommended minimum level of available phosphorus.

⁸ Oil levels can be increased to 2.0% in starter diets when given as a mash to control dust and increase feed palatability.

⁹ Avoid excessive body weight gain after 12 weeks.

Transition Period from Grow to Peak Egg Production

Frequently formulate to changing feed consumption during transition period until feed consumption is consistent.



Pre-Lay Ration

- Plan to feed for a maximum of 10–14 days before point of lay.
- Feed when most pullets show reddening of combs.
- It is important to increase medullary bone reserves.
- Begin introducing large particle calcium in Pre-Lay Diet.
- Discontinue pre-lay feeding with the commencement of egg production.

Transition Period

- Avoid excessive weight gain during the transition period.
- Body weight gain from 18–25 weeks should not exceed 20%.
- Occurring during the transition period:
 - Rapidly increasing egg production
 - Increasing egg size
 - Increasing body weight
- Feed consumption may increase slowly during transition:
 - In underweight birds
 - In flocks lacking uniformity
 - During high environmental temperatures
- Poor uniformity prolongs the transition period and may result in low peak and poor persistency of egg production.
- Monitor feed intake carefully during transition and adjust dietary nutrient concentration according to actual feed intakes.

Pre-Peaking Ration

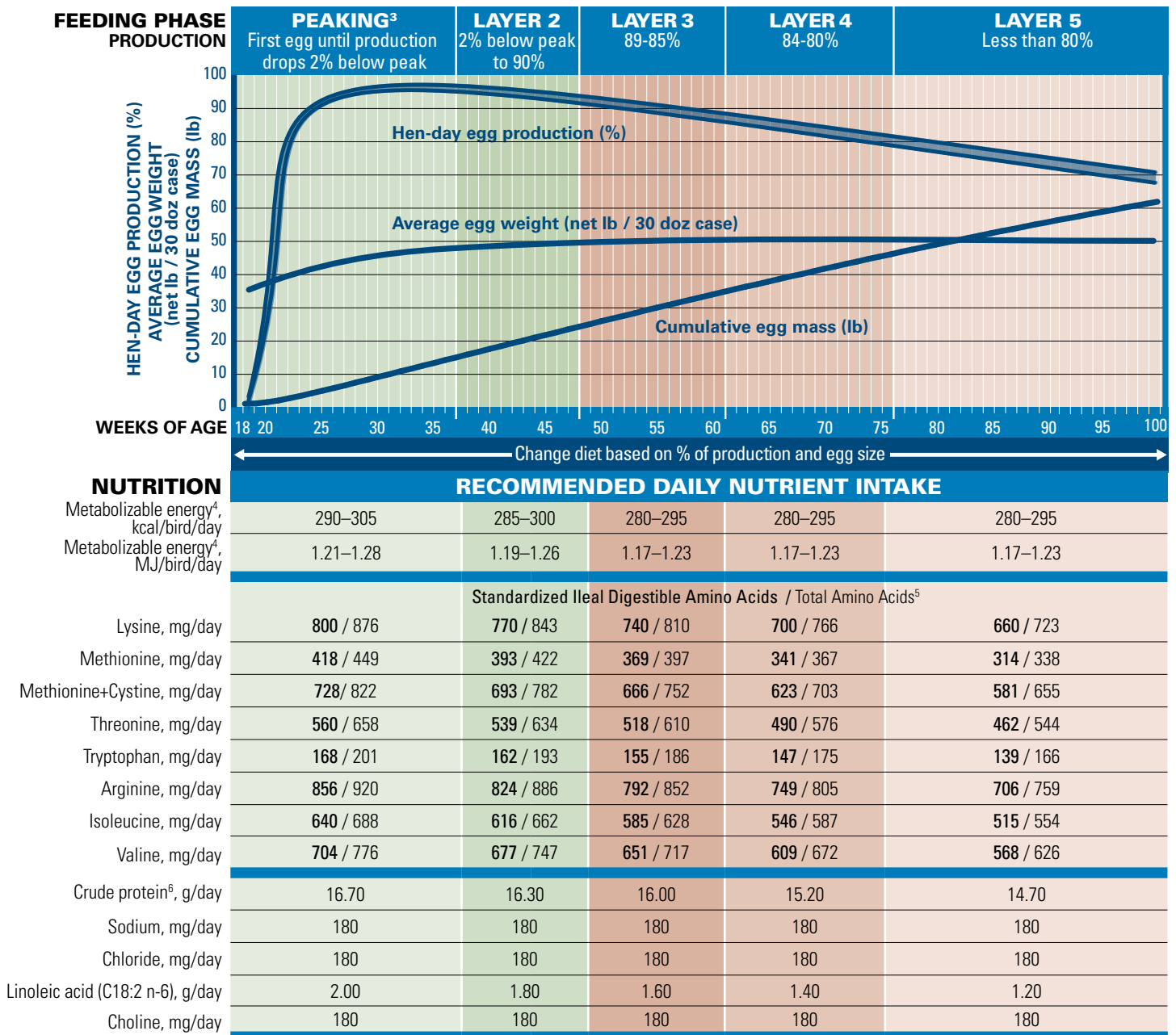
- Formulations for low feed intakes (17.6–18.7 lb/day per 100 birds) should be given as the flock enters egg production to better meet nutrient requirements.
- Begin Pre-Peaking Diet with onset of lay (1% egg production).
- Pre-Peaking Diet is given until average feed consumption reaches 20.9 lb/day per 100 birds.

Peaking Ration

- Birds should continue to grow during peaking period. Poor nutrition during this period can lead to loss of body weight and soft bones.
- Feed intake may be reduced if birds are not accustomed to extra large particle calcium (i.e. not using a Pre-Lay Diet).
- Monitor keel bone development during the peaking period. For more information on keel bone scoring, see the "[Understanding the Role of the Skeleton in Egg Production](#)" technical update at www.hyllinena.com.

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Production Period Nutritional Recommendations^{1,2}



CALCIUM AND PHOSPHORUS

	Calcium ^{7,8} g/day	Phosphorus (available) ^{7,9} mg/day	Calcium Particle Size (fine : coarse) (see p. 16)
First Egg–Peak	3.80	490	50% : 50%
Peak–35 Weeks	4.15	480	50% : 50%
Weeks 36–55	4.30	465	40% : 60%
Weeks 56–72	4.40	440	30% : 70%
Weeks 73–85	4.60	400	30% : 70%
Weeks 86+	4.75	380	30% : 70%

¹ All nutrient requirements are based on the feed ingredient table at the back of this guide.

² Crude protein, methionine+cystine, fat, linoleic acid, and / or energy may be changed to optimize egg size.

³ Peaking nutrient levels are calculated for birds at peak egg production. Prior to achieving peak egg production, the nutrient requirements will be lower.

⁴ A good approximation of the influence of temperature on energy needs is that for each 1°F change higher or lower than 72°F, subtract or add about 1.8 kcal / bird / day, respectively.

⁵ Recommendation for Total Amino Acids is only appropriate to corn and soybean meal diet. Where diets utilize other ingredients, recommendations for Standardized Ileal Digestible Amino Acids must be followed.

⁶ Diets should always be formulated to provide required intake of amino acid. Concentration of crude protein in diet will vary with raw material used. Crude protein value provided is an estimated typical value only.

⁷ Calcium and available phosphorus requirements are determined by flock age. When production remains higher and diets are fed for longer than ages shown, it is recommended to increase to calcium and phosphorus concentrations of next feeding phase.

⁸ Calcium carbonate particle size recommendation varies throughout lay. Refer to Calcium Particle Size Table. Dietary calcium levels may need to be adjusted based on limestone solubility.

⁹ Where other phosphorus systems are used, diets should contain recommended minimum level of available phosphorus.

Dietary Nutrient Concentrations for Production Period^{1,2} (According to Phase and Feed Intake)

FEEDING PHASE PRODUCTION	PEAKING ³ First egg until production drops 2% below peak					LAYER 2 2% below peak to 90%					LAYER 3 89–85%					LAYER 4 84–80%					LAYER 5 Less than 80%				
	RECOMMENDED CONCENTRATION																								
Metabolizable energy ⁴ , kcal/bird/day	290–305					285–300					290–305					290–305					290–305				
Metabolizable energy ⁴ , MJ/bird/day	1.21–1.28					1.19–1.26					1.17–1.23					1.17–1.23					1.17–1.23				
FEED CONSUMPTION (*Typical Feed Consumption)																									
g/day per bird	85	90	95*	100	105	90	95	100*	105	110	90	95	100*	105	110	90	95	100*	105	110	90	95	100*	105	110
lb/day per 100 birds	18.7	19.8	20.9*	22.0	23.1	19.8	20.9	22.0*	23.1	24.3	19.8	20.9	22.0*	23.1	24.3	19.8	20.9	22.0*	23.1	24.3	19.8	20.9	22.0*	23.1	24.3
Standardized Ileal Digestible Amino Acids																									
Lysine, %	0.94	0.89	0.84	0.80	0.76	0.86	0.81	0.77	0.73	0.70	0.82	0.78	0.74	0.70	0.67	0.78	0.74	0.70	0.67	0.64	0.73	0.69	0.66	0.63	0.60
Methionine, %	0.49	0.46	0.44	0.42	0.40	0.44	0.41	0.39	0.37	0.36	0.41	0.39	0.37	0.35	0.34	0.38	0.36	0.34	0.32	0.31	0.35	0.33	0.31	0.30	0.29
Methionine+Cystine, %	0.86	0.81	0.77	0.73	0.69	0.77	0.73	0.69	0.66	0.63	0.74	0.70	0.67	0.63	0.61	0.69	0.66	0.62	0.59	0.57	0.65	0.61	0.58	0.55	0.53
Threonine, %	0.66	0.62	0.59	0.56	0.53	0.60	0.57	0.54	0.51	0.49	0.58	0.55	0.52	0.49	0.47	0.54	0.52	0.49	0.47	0.45	0.51	0.49	0.46	0.44	0.42
Tryptophan, %	0.20	0.19	0.18	0.17	0.16	0.18	0.17	0.16	0.15	0.15	0.17	0.16	0.16	0.15	0.14	0.16	0.15	0.15	0.14	0.13	0.15	0.15	0.14	0.13	0.13
Arginine, %	1.01	0.95	0.90	0.86	0.82	0.92	0.87	0.82	0.78	0.75	0.88	0.83	0.79	0.75	0.72	0.83	0.79	0.75	0.71	0.68	0.78	0.74	0.71	0.67	0.64
Isoleucine, %	0.75	0.71	0.67	0.64	0.61	0.68	0.65	0.62	0.59	0.56	0.65	0.62	0.59	0.56	0.53	0.61	0.57	0.55	0.52	0.50	0.57	0.54	0.52	0.49	0.47
Valine, %	0.83	0.78	0.74	0.70	0.67	0.75	0.71	0.68	0.64	0.62	0.72	0.69	0.65	0.62	0.59	0.68	0.64	0.61	0.58	0.55	0.63	0.60	0.57	0.54	0.52
Total Amino Acids ⁵																									
Lysine, %	1.03	0.97	0.92	0.88	0.83	0.94	0.89	0.84	0.80	0.77	0.90	0.85	0.81	0.77	0.74	0.85	0.81	0.77	0.73	0.70	0.80	0.76	0.72	0.69	0.66
Methionine, %	0.53	0.50	0.47	0.45	0.43	0.47	0.44	0.42	0.40	0.38	0.44	0.42	0.40	0.38	0.36	0.41	0.39	0.37	0.35	0.33	0.38	0.36	0.34	0.32	0.31
Methionine+Cystine, %	0.97	0.91	0.87	0.82	0.78	0.87	0.82	0.78	0.74	0.71	0.84	0.79	0.75	0.72	0.68	0.78	0.74	0.70	0.67	0.64	0.73	0.69	0.66	0.62	0.60
Threonine, %	0.77	0.73	0.69	0.66	0.63	0.70	0.67	0.63	0.60	0.58	0.68	0.64	0.61	0.58	0.55	0.64	0.61	0.58	0.55	0.52	0.60	0.57	0.54	0.52	0.49
Tryptophan, %	0.24	0.22	0.21	0.20	0.19	0.21	0.20	0.19	0.18	0.18	0.21	0.20	0.19	0.18	0.17	0.19	0.18	0.18	0.17	0.16	0.18	0.17	0.17	0.16	0.15
Arginine, %	1.08	1.02	0.97	0.92	0.88	0.98	0.93	0.89	0.84	0.81	0.95	0.90	0.85	0.81	0.77	0.89	0.85	0.81	0.77	0.73	0.84	0.80	0.76	0.72	0.69
Isoleucine, %	0.81	0.76	0.72	0.69	0.66	0.74	0.70	0.66	0.63	0.60	0.70	0.66	0.63	0.60	0.57	0.65	0.62	0.59	0.56	0.53	0.62	0.58	0.55	0.53	0.50
Valine, %	0.91	0.86	0.82	0.78	0.74	0.83	0.79	0.75	0.71	0.68	0.80	0.75	0.72	0.68	0.65	0.75	0.71	0.67	0.64	0.61	0.70	0.66	0.63	0.60	0.57
Crude protein ⁶ , %	19.65	18.56	17.58	16.70	15.90	18.11	17.16	16.30	15.52	14.82	17.78	16.84	16.00	15.24	14.55	16.89	16.00	15.20	14.48	13.82	16.33	15.47	14.70	14.00	13.36
Sodium, %	0.21	0.20	0.19	0.18	0.17	0.20	0.19	0.18	0.17	0.16	0.20	0.19	0.18	0.17	0.16	0.20	0.19	0.18	0.17	0.16	0.20	0.19	0.18	0.17	0.16
Chloride, %	0.21	0.20	0.19	0.18	0.17	0.20	0.19	0.18	0.17	0.16	0.20	0.19	0.18	0.17	0.16	0.20	0.19	0.18	0.17	0.16	0.20	0.19	0.18	0.17	0.16
Linoleic acid (C18:2 n-6), %	2.35	2.22	2.11	2.00	1.90	2.00	1.89	1.80	1.71	1.64	1.78	1.68	1.60	1.52	1.45	1.56	1.47	1.40	1.33	1.27	1.33	1.26	1.20	1.14	1.09
Choline, mg/kg	2118	2000	1895	1800	1714	2000	1895	1800	1714	1636	2000	1895	1800	1714	1636	2000	1895	1800	1714	1636	2000	1895	1800	1714	1636
CALCIUM AND PHOSPHORUS CHANGES BASED ON AGE																									
	Peak–35 Weeks					Weeks 36–55					Weeks 56–72					Weeks 73–85					Weeks 86+				
Calcium ^{7,8} , %	4.88	4.61	4.37	4.15	3.95	4.78	4.53	4.30	4.10	3.91	4.89	4.63	4.40	4.19	4.00	5.11	4.84	4.60	4.38	4.18	5.28	5.00	4.75	4.52	4.32
Phosphorus (available) ^{7,9} , %	0.56	0.53	0.51	0.48	0.46	0.52	0.49	0.47	0.44	0.42	0.49	0.46	0.44	0.42	0.40	0.44	0.42	0.40	0.38	0.36	0.42	0.40	0.38	0.36	0.35

¹ All nutrient requirements are based on the feed ingredient table at the back of this guide.

² Crude protein, methionine+cystine, fat, linoleic acid, and / or energy may be changed to optimize egg size.

³ Peaking nutrient levels are calculated for birds at peak egg production. Prior to achieving peak egg production, the nutrient requirements will be lower.

⁴ A good approximation of the influence of temperature on energy needs is that for each 1°F change higher or lower than 72°F, subtract or add about 1.8 kcal /bird / day, respectively.

⁵ Recommendation for Total Amino Acids is only appropriate to corn and soybean meal diet. Where diets utilize other ingredients, recommendations for Standardized Ileal Digestible Amino Acids must be followed.

⁶ Diets should always be formulated to provide required intake of amino acid. Concentration of crude protein in diet will vary with raw material used. Crude protein value provided is an estimated typical value only.

⁷ Calcium and available phosphorus requirements are determined by flock age. When production remains higher and diets are fed for longer than ages shown, it is recommended to increase to calcium and phosphorus concentrations of next feeding phase.

⁸ Calcium carbonate particle size recommendation varies throughout lay. Refer to Calcium Particle Size Table. Dietary calcium levels may need to be adjusted based on limestone solubility.

⁹ Where other phosphorus systems are used, diets should contain recommended minimum level of available phosphorus.

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Non-Fasting Molt Recommendations

Decision to molt is based on:

- Projected feed costs
- Egg prices
- Differential prices between egg sizes
- Cost of pullet replacement
- Value of hens at end of lay
- Flock performance

Non-Fasting Molt

- Hy-Line laying hens will perform very well after a rest.
- The optimum age for molting is usually between 65 (early) to 75 (late) weeks of age.
- Induced molting can extend the productive life of a flock by improving rate of lay, shell quality, and albumen height.
- Post-molt performance will be lower than best pre-molt values.
- Egg size will essentially remain unaffected and will continue to increase after egg production resumes.
- Birds need free access to water at all times.

- High sodium levels in drinking water (i.e., 100 ppm or higher) can adversely affect non-fasting molt.
- Best post-molt egg production is achieved after complete cessation of egg production that lasts for at least 2 weeks.
- The goal is to maintain 18 week body weight (2.78–2.87 lb) during rest period.
- Reducing body weight to 18 week body weight (approximately 23% reduction) will result in best post-molt performance.
- After initial body weight loss, body weight can be held steady by a combination of adjusting the number of feedings per day and / or a shift to a higher-energy (laying-hen-type) diet.
- Closely monitor body weight of flock during molt process.
- Body weights should be collected twice per week, from the same cages every time.
- Cages should be selected from bottom, middle and top tiers; all rows; and from front, middle and end of house.

Non-Fasting Molt Program

MOLT DAY	LIGHT (hours per day)	FEED TYPE	FEED MODIFICATION ¹	FEED INTAKE ² (g / day per bird)	HOUSE TEMPERATURE ³ (°F)	COMMENTS
-7 to -5	16	Layer diet	Fine-particle CaCO ₃	Full feed	75–77	Remove large-particle size CaCO ₃ and replace with fine-particle CaCO ₃ (less than 2 mm mean diameter). Do NOT change percent calcium in laying-hen diet
-4 to -1	24	Layer diet	Fine-particle CaCO ₃ , no added salt (NaCl)	Full feed	75–77	
0–6	6–8 ⁴	Molt diet ⁵	Fine-particle CaCO ₃	54–64	81–82	Higher house temperatures will reduce feed intake and facilitate reduction in body weight to 18 week target weight (Note: white laying hens should not lose more than 25% of the pre-molt body weight)
7–17	6–8	Molt diet	—	54–64	81–82	Maintain body weight
18–19	12 or 16 ⁶	Layer diet ⁷	Mixture of fine- and coarse-particle CaCO ₃ as in a normal layer diet	64–73	81–82	Control (limit) feed intake to avoid fat birds
20–21	16 ⁶	Layer diet ⁷	—	Full feed	79–81	Lower house temperature as needed to increase feed intake
22–24	16	Layer diet ⁷	—	Full feed ⁷	75–77	Lower ambient temperature to “normal”

¹ May include a probiotic or prebiotic through all stages of molt program.

² Feed intake depends on house temperature. Lower temperatures (colder) may require more feed.

³ Monitor air quality in house. Suggested house temperatures may not be achievable in cold weather.

⁴ Set lights at 8 hours or natural day length in open-sided houses. Normally not necessary to change light intensity.

⁵ Molt Diet is high in fiber (low in energy) and contains no added sodium (Na) (i.e., no added NaCl or NaHCO₃).

⁶ Light stimulate birds into production by increasing light hours to number of hours they were given before molt (i.e., 15 or 16 hours). Increase can be performed in a single day (i.e., from 8 hours to 16 hours in a single day) or two weekly steps (i.e., from 8 to 12 hours and then from 12 to 16 hours). Monitor and control feed intake for first few days after light stimulation to avoid fat birds as they return into lay (which would significantly increase egg weight in second cycle).

⁷ According to post-molt nutrition recommendations.

Molt Diet Recommendations

NUTRITION	RECOMMENDED NUTRIENT CONCENTRATION
Metabolizable energy ¹ , kcal/lb	1000–1270
Metabolizable energy ¹ , kcal/kg	2205–2800
Metabolizable energy ¹ , MJ/kg	9.32–11.72
Standardized Ileal Digestible Amino Acids / Total Amino Acids ²	
Lysine, %	0.30 / 0.33
Methionine, %	0.15 / 0.16
Methionine+Cystine, %	0.32 / 0.36
Threonine, %	0.18 / 0.21
Tryptophan, %	0.10 / 0.12
Arginine, %	0.38 / 0.41
Isoleucine, %	0.18 / 0.20
Valine, %	0.23 / 0.26
Crude protein ³ , %	8.50
Calcium ⁴ , %	1.3–2.0
Phosphorus (available), %	0.25
Sodium ⁵ , %	0.03
Chloride, %	0.03

¹ Recommended energy range is based on energy values shown in feed ingredient table at back of this guide. It is important that target concentrations of dietary energy are adjusted according to energy system applied to raw material matrix if values differ from those referred for raw materials in this guide.

² Recommendation for Total Amino Acids is only appropriate to corn and soybean meal diet. Where diets utilize other ingredients, recommendations for Standardized Ileal Digestible Amino Acids must be followed.

³ Diets should always be formulated to provide required intake of amino acid. Concentration of crude protein in diet will vary with raw material used. Crude protein value provided is an estimated typical value only.

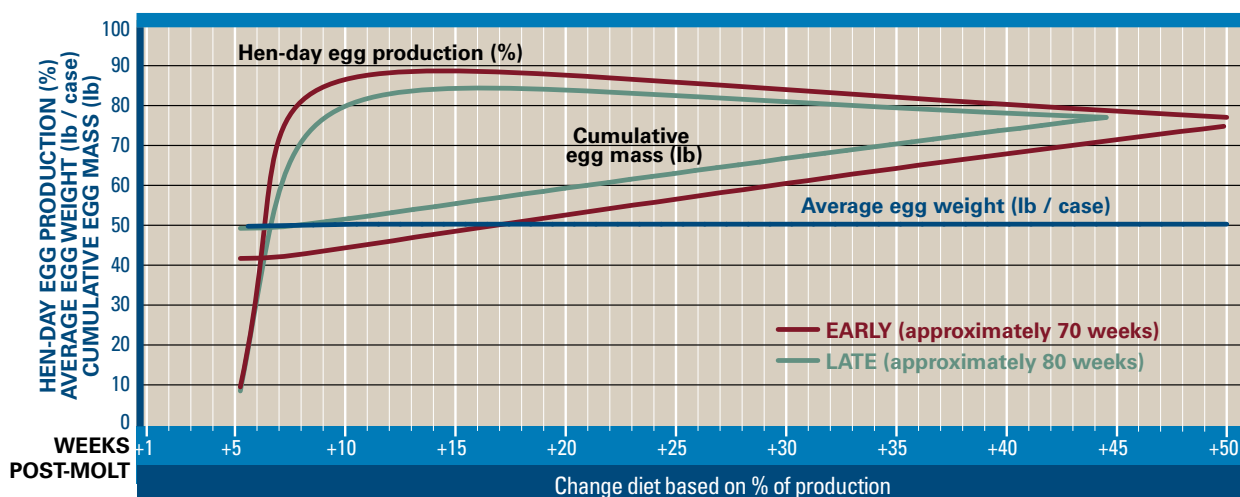
⁴ Calcium carbonate should be in particle sizes of less than 2 mm.

⁵ Sodium content in Molt Diet should not exceed 0.035%.



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Post-Molt Nutrition Recommendations¹



NUTRITION

RECOMMENDED NUTRIENT INTAKE

Metabolizable energy ² , kcal/bird/day	285 / 300	280 / 295	280 / 295	280 / 295
Metabolizable energy ² , MJ/bird/day	1.19 / 1.26	1.17 / 1.23	1.17 / 1.23	1.17 / 1.23
Standardized Ileal Digestible Amino Acids / Total Amino Acids ³				
Lysine, mg/day	760 / 832	730 / 799	700 / 766	660 / 723
Methionine, mg/day	397 / 427	373 / 401	349 / 375	322 / 346
Methionine+Cystine, mg/day	692 / 780	657 / 741	630 / 711	587 / 662
Threonine, mg/day	532 / 626	511 / 601	490 / 576	462 / 544
Tryptophan, mg/day	160 / 191	154 / 184	147 / 176	139 / 166
Arginine, mg/day	813 / 874	781 / 840	749 / 805	706 / 759
Isoleucine, mg/day	608 / 654	584 / 628	553 / 595	515 / 554
Valine, mg/day	668 / 737	642 / 708	616 / 679	574 / 633
Crude protein ⁴ , g/day	15.75	15.25	15.00	14.75
Sodium, mg/day	180	180	180	180
Chloride, mg/day	180	180	180	180
Linoleic acid (C18:2 n-6), g/day	1.60	1.40	1.20	1.00
Choline, mg/day	180.00	180.00	180.00	180.00

RECOMMENDED POST-MOLT DIETARY CALCIUM AND PHOSPHORUS CHANGES BASED ON AGE

	1–15 Weeks Post Molt	16–24 Weeks Post Molt	25–33 Weeks Post Molt	33–50 Weeks Post Molt
Calcium ⁵ , g/day	4.35	4.50	4.60	4.80
Phosphorus (available) ⁶ , mg/day	490	475	450	410
Calcium Particle Size (fine:coarse)	40% : 60%	35% : 65%	30% : 70%	30% : 70%

¹ Crude protein, methionine+cystine, fat, linoleic acid, and / or energy may be changed to optimize egg size.

² A good approximation of the influence of temperature on energy needs is that for each 1°F change higher or lower than 72°F, subtract or add about 1.8 kcal /bird /day, respectively.

³ Recommendation for Total Amino Acids is only appropriate to corn and soybean meal diet. Where diets utilize other ingredients, recommendations for Standardized Ileal Digestible Amino Acids must be followed.

⁴ Diets should always be formulated to provide required intake of amino acid. Concentration of crude protein in diet will vary with raw material used. Crude protein value provided is an estimated typical value only.

⁵ 65% of calcium carbonate should be in 2–4 mm particle size.

⁶ Where other phosphorus systems are used, diets should contain recommended minimum level of available phosphorus.

Disease Control

A flock of pullets or layers can only perform up to its genetic potential when disease influence is minimized. The diseases of economic importance vary widely between locations, but in every case the challenge is to identify and control those diseases.

Biosecurity

Biosecurity is the best method of avoiding diseases. A good biosecurity program identifies and controls the most likely ways a disease could enter the farm.

- Human and equipment movement onto the farm should be strictly controlled.
- Visitors to the farm should be limited to those essential for its operation.
- Visits should be documented in a logbook.
- All visitors and workers should shower at a central location before entering.
- Clean boots, clothing and head cover should be provided for workers and visitors.
- Clean footbaths containing disinfectant should be placed outside entries to all poultry houses.
- If possible, avoid using outside crews or equipment for vaccination, moving, and beak trimming.
- Ideally, workers should be limited to a single house.
- For those visiting a number of flocks, flocks visited on one day should be limited. Always progress from younger to older and from healthy to sick flocks. After visiting a sick flock, no other houses should be entered.
- Removal of flocks from the farm is an opportunity for disease to be introduced, as trucks and crews have often been on other farms.
- A single-aged growing farm using an all-in, all-out principle is best to prevent transmission of disease from older flocks to younger, susceptible flocks.
- Houses should be designed to prevent exposure to wild birds, insects and rodents.
- Quickly and properly dispose of dead chickens.

Rodents

Rodents are known carriers of many poultry diseases and the most common reason for re-contamination of a cleaned and disinfected poultry facility. They are also responsible for house-to-house spread of disease on a farm.

- The farm should be free of debris and tall grass that provide a hiding area for rodents.
- The perimeter of each house should have a 1 m wide area of crushed rock or concrete to prevent rodents from burrowing into the house.
- Feed and eggs should be stored in rodent-proof areas.
- Bait stations should be placed throughout house and maintained with fresh rodenticide.

Cleaning and Disinfection

Cleaning and disinfection of the house between flocks reduces infection pressure for the next flock.

- Allow a minimum of 2 weeks downtime between flocks.
- All feed and manure should be removed from the house before cleaning.
- Thoroughly clean air inlets, fan housing, fan blades and fan louvers.
- Heating the house during washing improves the removal of organic matter.

- The house should be cleaned of organic matter with a high-pressure spray of warm water.
- Use foam / gel detergent to soak into organic matter and equipment.
- Wash the upper portion of the house before the pit.
- Use high pressure warm water to rinse.
- Allow the house to dry.
- After it is fully dry, apply foam / spray disinfectant followed by fumigation.
- Flush and sanitize water lines.
- The monitoring of poultry houses for the presence of *Salmonella*, particularly *Salmonella* Enteritidis, by routine environmental testing is recommended.
- Allow the house to dry before repopulating.

Vertically Transmitted Diseases

- Some diseases are known to be transmitted from infected breeders to progeny.
- Disease-free breeders are the first step in control of these diseases for commercial layers.
- All breeders directly under Hy-Line International's control are free of *Mycoplasma gallisepticum*, *Mycoplasma synoviae*, *Salmonella* Pullorum, *Salmonella* Gallinarum, *Salmonella* Enteritidis, *Salmonella* Typhimurium and lymphoid leukosis.
- Due to the possibility of horizontal transmission of these diseases, later generations may not remain free.
- It is the responsibility of breeding and commercial flock owners to prevent horizontal transmission of these diseases and to continue testing to be assured of a negative status.



COCCIDIA

This parasitic infection of the intestines may lead to gut damage and, in severe infestations, death. More commonly, poor control of sub-clinical infection reduces feed conversion or leaves pullets with chronic, irreversible gut damage. Pullet flocks may be uneven or underweight at housing and not perform to their full potential in lay. Control of coccidia includes the following measures (check local regulations):

- Use ionophores or chemicals on a step-down program to ensure immunity in pullets.
- Live vaccine use is an alternative to anti-coccidial drug treatments.
- Live vaccines are available that can be administered by spray in the hatchery or by feed or water application during the first few days in the brooder house.
- Control of flies and beetles, which are vectors of coccidia spread.
- Thorough cleaning and disinfection of houses reduces challenge pressure.
- Limit bird access to manure belts.
- Cocci vaccines require cycling; discuss this with the vaccine manufacturer.

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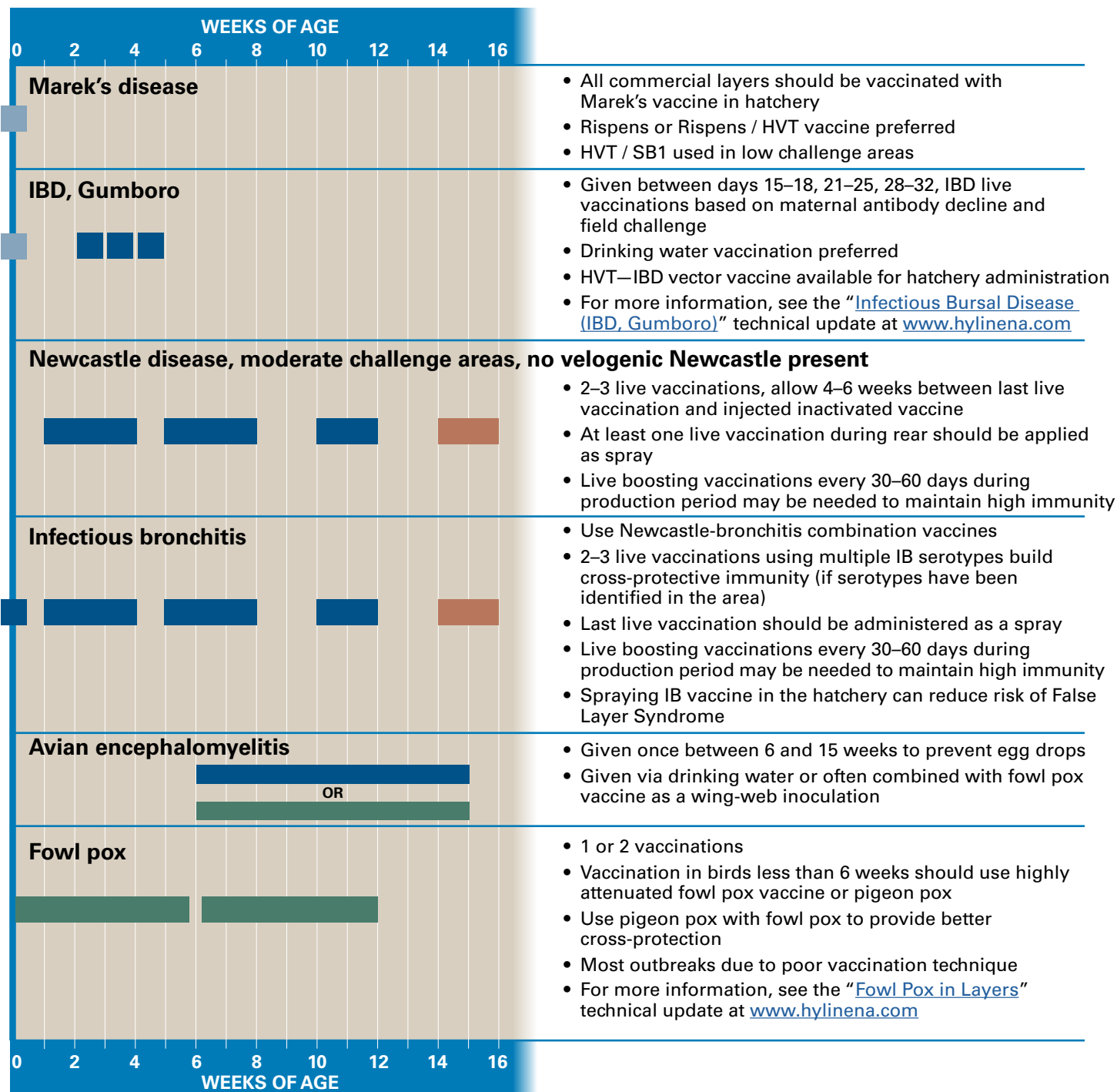
Vaccination Recommendations

Vaccination

Certain diseases are too widespread or difficult to eradicate and require a routine vaccination program. In general, all layer flocks should be vaccinated against Marek's disease, Newcastle disease (NDV), infectious bronchitis (IB), infectious bursal disease (IBD or Gumboro), avian encephalomyelitis (AE) and fowl pox. Other vaccinations are added to the program as local disease challenges dictate.

A single program cannot be recommended for all regions. Follow label instructions provided by the vaccine manufacturer. Use only approved vaccines. Consult with local veterinarians to determine the best vaccination program for your area.

BASIC COMMERCIAL LAYER VACCINE APPLICATIONS



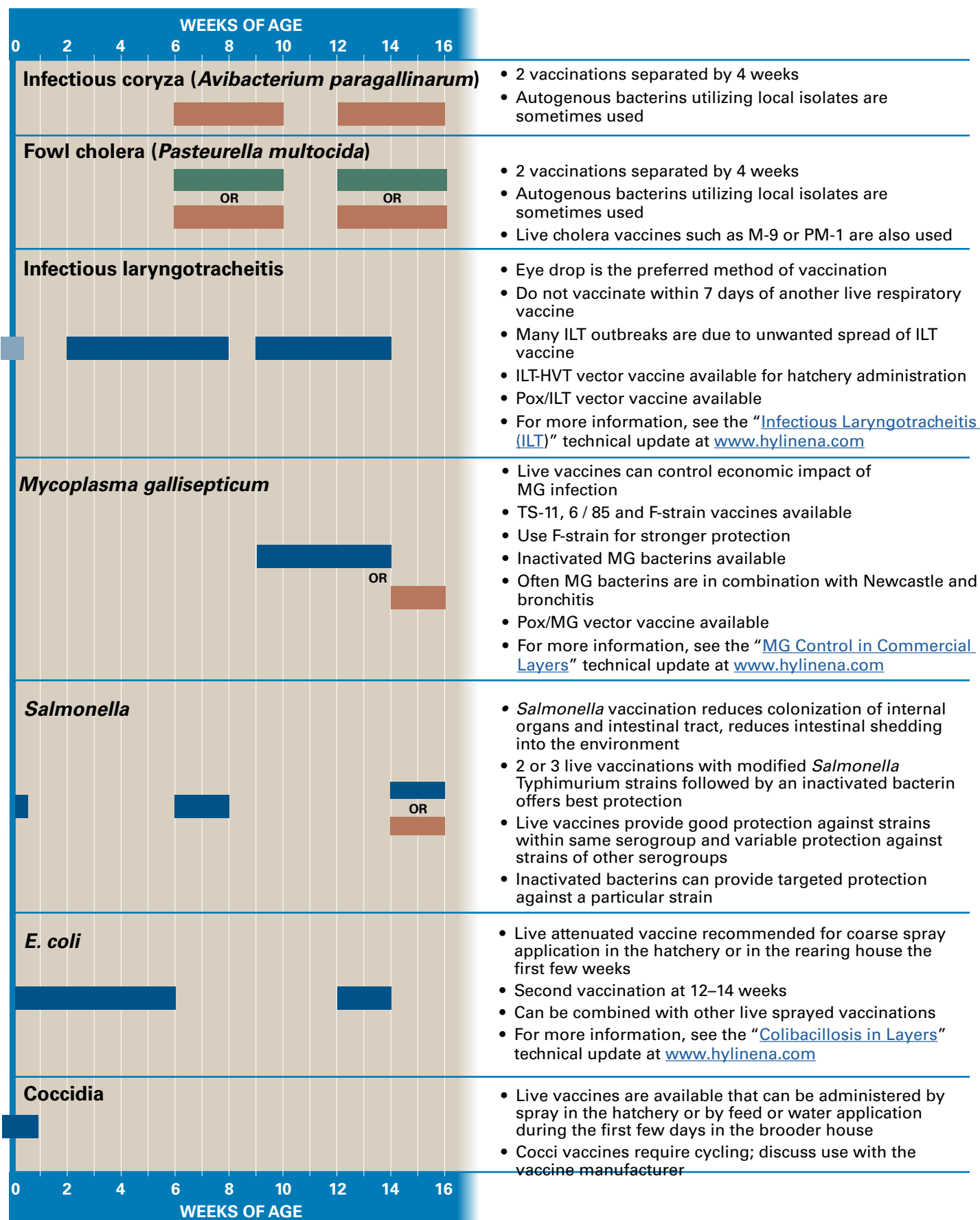
Live hatchery vaccines, given subcutaneous
 Live vaccines, administered by drinking water, spray or eyedrop

Live vaccines, administered via wing-web inoculation
 Inactivated vaccines, injected via intramuscular or subcutaneous route

Vaccination Recommendations *(continued)*

OPTIONAL COMMERCIAL LAYER VACCINE APPLICATIONS

Use if these diseases are prevalent in the area. Follow label instructions provided by the vaccine manufacturer. Use only approved vaccines. Consult a local veterinarian for advice in designing an effective vaccination program for your farm.



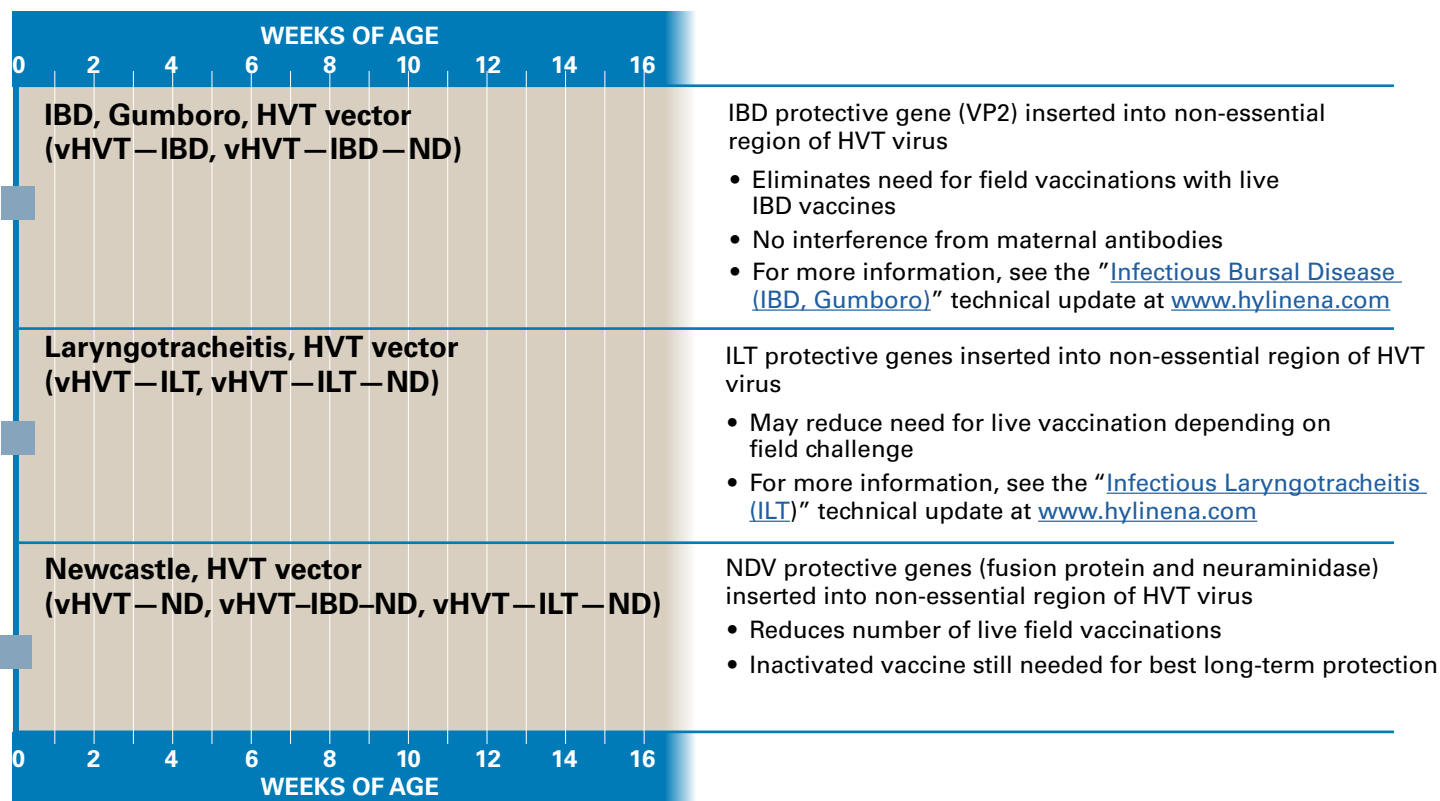
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Vaccination Recommendations *(continued)*

RECOMBINANT HVT (vHVT) VACCINES

Vaccines using recombinant vector technology offer the convenience of hatchery administration with no adverse effects caused by some live field vaccinations. For best Marek's disease protection use Rispens vaccine in combination with recombinant HVT vaccine.

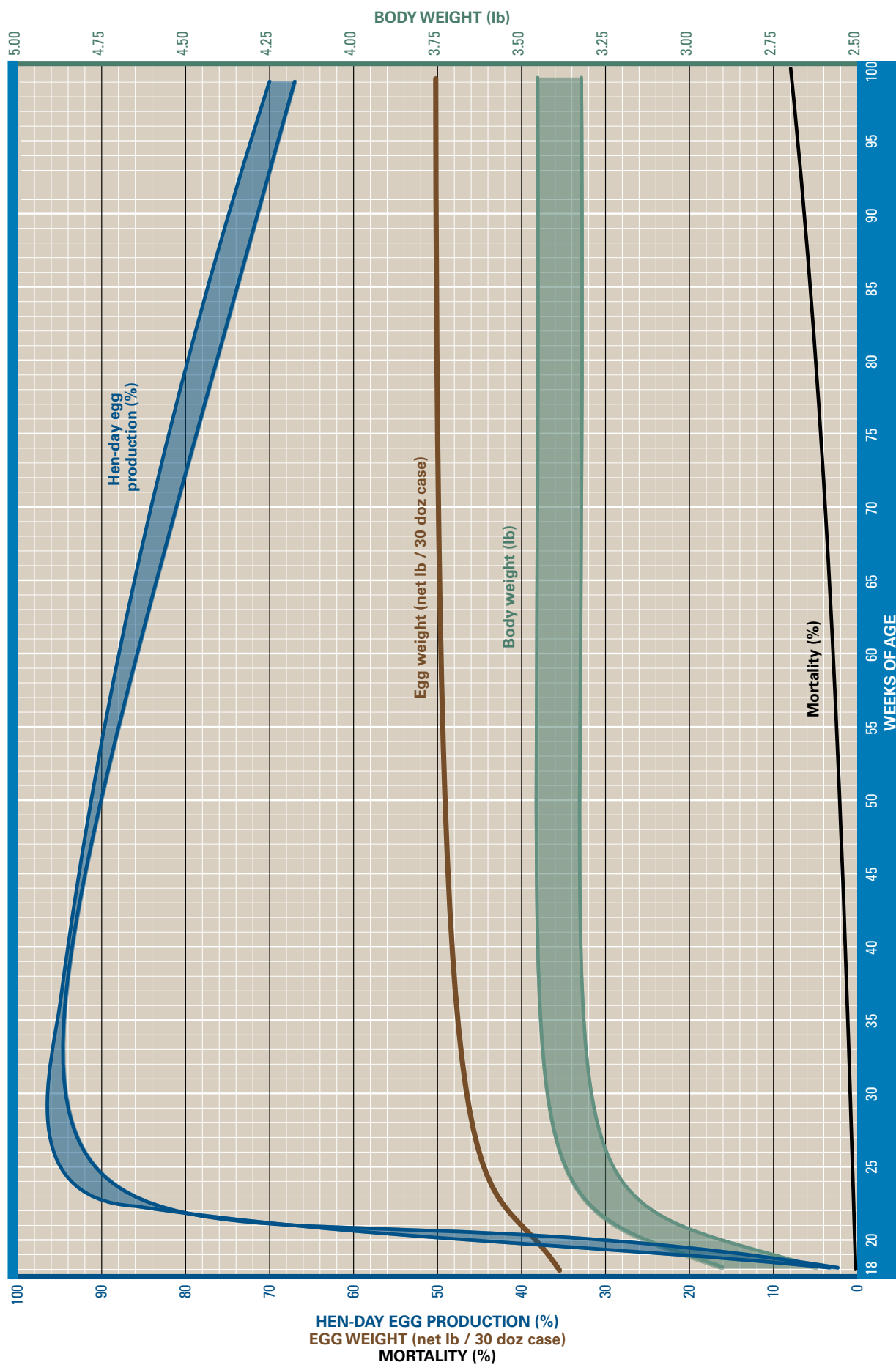
CAUTION: Do not use another HVT vaccine when using HVT-vectored vaccines; do not use multiple vectored vaccines unless combined by the manufacturer.



Live hatchery vaccines, given subcutaneous
 Live vaccines, administered by drinking water, spray or eyedrop

Live vaccines, administered via wing-web inoculation
 Inactivated vaccines, injected via intramuscular or subcutaneous route

Performance Graph



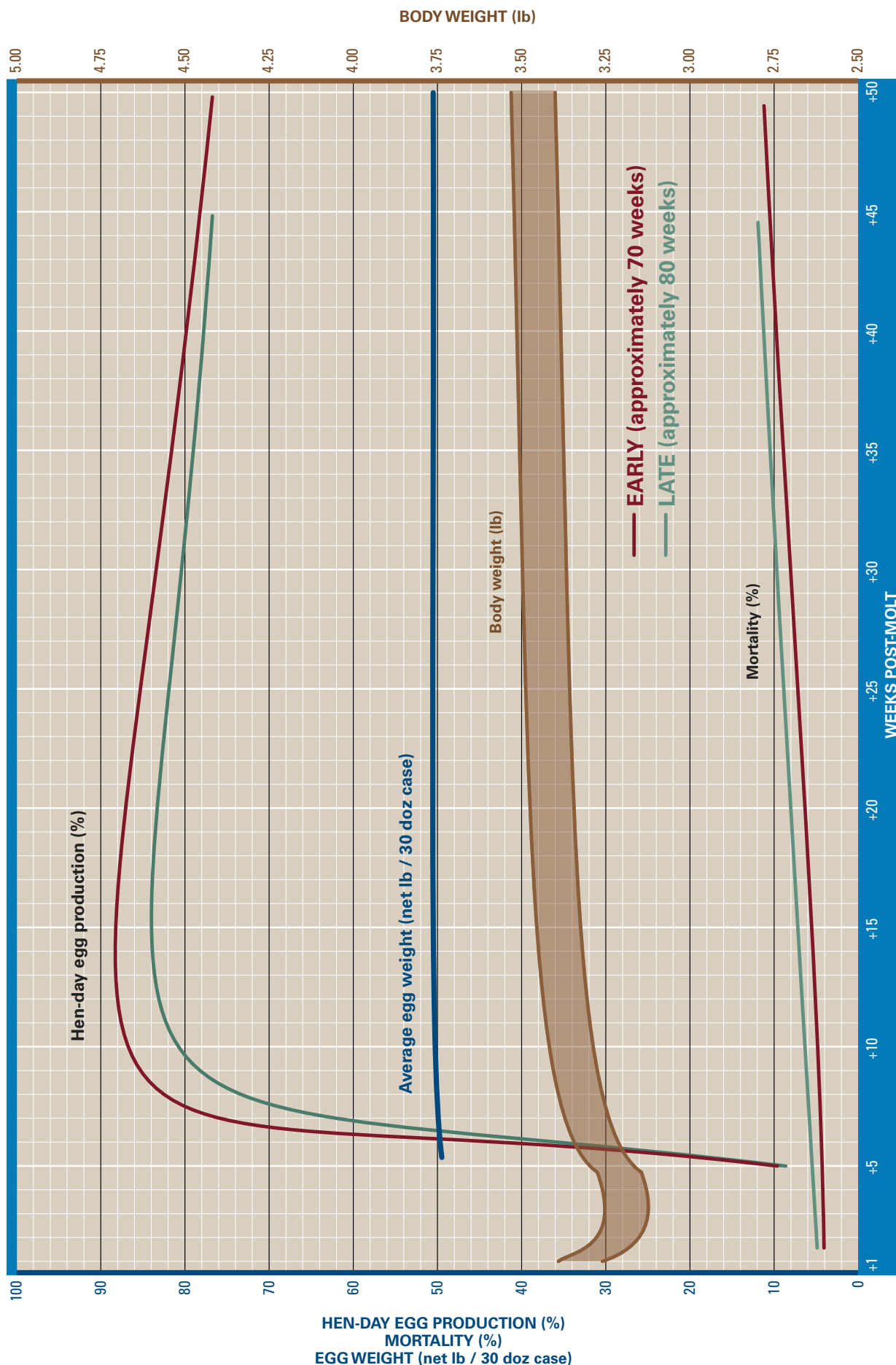
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Post-Molt Performance Table

WEEKS POST MOLT	% HEN- DAY		HEN-DAY EGGS Cumulative		HEN-HOUSED EGGS Weekly		HEN-HOUSED EGGS Cumulative		MORT- ALITY (%)		BODY WEIGHT (lb)		FEED INTAKE (lb/100)		HEN-HOUSED EGG MASS Cumulative (lb)		AVERAGE EGG WEIGHT (lb/case)
	Early	Late	Early	Late	Early	Late	Early	Late	Early	Late	Low	High	Low	High	Early	Late	
+1	—	—	319.3	375.8	—	—	313.5	367.6	3.9	4.7	3.26	3.39	10.4	11.0	—	—	—
+2	—	—	319.3	375.8	—	—	313.5	367.6	4.0	4.9	3.13	3.26	10.4	11.0	—	—	—
+3	—	—	319.3	375.8	—	—	313.5	367.6	4.0	5.0	3.13	3.26	14.1	14.8	—	—	—
+4	—	—	319.3	375.8	—	—	313.5	367.6	4.1	5.2	3.13	3.26	17.2	17.9	—	—	—
+5	10	9	320.0	376.4	0.7	0.6	314.2	368.2	4.2	5.3	3.15	3.28	18.7	19.4	41.5	49.0	50.0
+6	48	40	323.3	379.2	3.2	2.6	317.4	370.8	4.3	5.5	3.22	3.35	19.8	20.5	41.6	49.1	50.1
+7	73	61	328.4	383.5	4.9	4.0	322.3	374.9	4.3	5.6	3.26	3.39	20.9	21.6	42.0	49.4	50.2
+8	84	72	334.3	388.5	5.6	4.7	327.9	379.6	4.4	5.8	3.27	3.40	21.4	22.0	42.7	50.0	50.3
+9	87	79	340.4	394.0	5.8	5.2	333.7	384.8	4.6	6.0	3.28	3.41	21.8	22.5	43.5	50.7	50.4
+10	88	82	346.6	399.8	5.9	5.4	339.6	390.2	4.6	6.1	3.28	3.41	22.0	22.7	44.3	51.4	50.5
+11	88	83	352.7	405.6	5.9	5.4	345.5	395.6	4.7	6.3	3.29	3.42	22.0	22.7	45.1	52.1	50.5
+12	89	84	359.0	411.5	5.9	5.5	351.4	401.1	4.9	6.5	3.30	3.43	22.0	22.9	45.9	52.9	50.5
+13	89	85	365.2	417.4	5.9	5.6	357.3	406.7	5.0	6.7	3.31	3.44	22.0	22.9	46.8	53.7	50.5
+14	88	85	371.4	423.4	5.8	5.5	363.2	412.2	5.2	6.8	3.32	3.45	22.0	23.0	47.6	54.4	50.5
+15	87	84	377.4	429.2	5.8	5.5	368.9	417.7	5.3	7.0	3.32	3.45	22.0	23.0	48.4	55.2	50.5
+16	87	84	383.5	435.1	5.8	5.5	374.7	423.1	5.5	7.2	3.33	3.46	22.0	23.0	49.2	56.0	50.5
+17	87	83	389.6	440.9	5.7	5.4	380.4	428.5	5.6	7.4	3.34	3.47	22.0	23.0	50.0	56.7	50.5
+18	87	83	395.7	446.7	5.7	5.4	386.2	433.9	5.8	7.6	3.34	3.47	22.0	23.0	50.8	57.5	50.5
+19	87	83	401.8	452.6	5.7	5.4	391.9	439.3	6.0	7.8	3.34	3.47	22.0	23.0	51.6	58.2	50.5
+20	86	82	407.8	458.3	5.7	5.3	397.5	444.5	6.1	7.9	3.34	3.47	22.0	23.0	52.4	59.0	50.5
+21	86	82	413.8	464.0	5.6	5.3	403.2	449.8	6.3	8.0	3.35	3.48	22.0	23.0	53.2	59.7	50.5
+22	86	82	419.9	469.8	5.6	5.3	408.8	455.1	6.5	8.2	3.35	3.48	22.0	23.0	54.0	60.4	50.5
+23	86	82	425.9	475.5	5.6	5.3	414.4	460.4	6.7	8.3	3.35	3.48	22.0	23.0	54.8	61.2	50.5
+24	86	82	431.9	481.3	5.6	5.3	420.0	465.6	6.8	8.5	3.35	3.48	22.0	23.0	55.5	61.9	50.5
+25	86	82	437.9	487.0	5.6	5.2	425.6	470.9	7.0	8.6	3.35	3.48	22.0	23.0	56.3	62.6	50.5
+26	86	82	443.9	492.7	5.6	5.2	431.2	476.1	7.2	8.8	3.36	3.49	22.0	23.0	57.1	63.4	50.5
+27	86	82	450.0	498.5	5.6	5.2	436.8	481.3	7.4	8.9	3.36	3.49	22.0	23.0	57.9	64.1	50.5
+28	85	81	455.9	504.1	5.5	5.2	442.3	486.5	7.6	9.1	3.36	3.49	22.0	23.0	58.6	64.8	50.5
+29	85	81	461.9	509.8	5.5	5.1	447.8	491.6	7.8	9.2	3.36	3.49	22.0	23.0	59.4	65.5	50.5
+30	85	81	467.8	515.5	5.5	5.1	453.3	496.8	7.9	9.4	3.36	3.49	22.0	23.0	60.2	66.3	50.5
+31	85	81	473.8	521.2	5.5	5.1	458.7	501.9	8.0	9.5	3.37	3.50	22.0	23.0	60.9	67.0	50.5
+32	84	80	479.6	526.8	5.4	5.1	464.1	507.0	8.2	9.7	3.37	3.50	22.0	23.0	61.7	67.7	50.5
+33	84	80	485.5	532.4	5.4	5.1	469.5	512.0	8.3	9.8	3.37	3.50	22.0	23.0	62.5	68.4	50.5
+34	83	80	491.3	538.0	5.3	5.0	474.9	517.1	8.5	10.0	3.37	3.50	22.0	23.0	63.2	69.1	50.5
+35	82	80	497.1	543.6	5.2	5.0	480.1	522.1	8.6	10.1	3.37	3.50	22.0	23.0	63.9	69.8	50.5
+36	82	80	502.8	549.2	5.2	5.0	485.3	527.1	8.8	10.3	3.38	3.51	22.0	23.0	64.7	70.5	50.5
+37	81	79	508.5	554.7	5.2	5.0	490.5	532.1	8.9	10.4	3.38	3.51	22.0	23.0	65.4	71.2	50.5
+38	80	79	514.1	560.2	5.1	4.9	495.6	537.0	9.1	10.6	3.38	3.51	22.0	23.0	66.1	71.9	50.5
+39	80	79	519.7	565.7	5.1	4.9	500.7	542.0	9.2	10.7	3.38	3.51	22.0	23.0	66.8	72.6	50.5
+40	80	78	525.3	571.2	5.1	4.9	505.8	546.8	9.4	10.9	3.38	3.51	22.0	23.0	67.5	73.3	50.5
+41	79	78	530.8	576.7	5.0	4.9	510.8	551.7	9.5	11.0	3.39	3.52	22.0	23.0	68.2	73.9	50.5
+42	79	78	536.3	582.1	5.0	4.9	515.8	556.5	9.7	11.2	3.39	3.52	22.0	23.0	68.9	74.6	50.5
+43	79	78	541.9	587.6	5.0	4.8	520.7	561.4	9.8	11.3	3.39	3.52	22.0	23.0	69.6	75.3	50.5
+44	78	77	547.3	593.0	4.9	4.8	525.7	566.2	10.0	11.5	3.39	3.52	22.0	23.0	70.3	76.0	50.5
+45	78	77	552.8	598.4	4.9	4.8	530.6	570.9	10.1	11.6	3.39	3.52	22.0	23.0	71.0	76.6	50.5
+46	78	—	558.2	—	4.9	—	535.4	—	10.3	—	3.40	3.53	22.0	23	71.7	—	50.5
+47	78	—	563.7	—	4.9	—	540.3	—	10.4	—	3.40	3.53	22.0	23	72.4	—	50.5
+48	77	—	569.1	—	4.8	—	545.2	—	10.6	—	3.40	3.53	22.0	23	73.1	—	50.5
+49	77	—	574.5	—	4.8	—	550.0	—	10.7	—	3.40	3.53	22.0	23	73.7	—	50.5
+50	77	—	579.8	—	4.8	—	554.8	—	10.9	—	3.40	3.53	22.0	23	74.4	—	50.5

Early: A molting program starting at approximately 70 weeks of age.
Late: A molting program starting at approximately 80 weeks of age.

Post-Molt Performance Graph



Early: A molting program starting at approximately 70 weeks of age.
Late: A molting program starting at approximately 80 weeks of age.

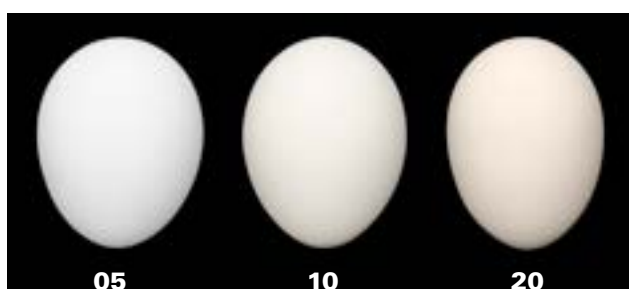
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Egg Standards and Egg Size Distribution

EGG QUALITY			
AGE (weeks)	HAUGH UNITS	% SOLIDS	BREAKING STRENGTH
18	98.0	22.4	4280
20	97.6	22.9	4260
22	96.8	23.2	4250
24	96.0	23.5	4240
26	95.3	23.7	4220
28	94.6	23.9	4200
30	93.9	24.1	4180
32	93.2	24.3	4160
34	92.6	24.4	4140
36	92.0	24.5	4120
38	91.4	24.6	4110
40	90.8	24.6	4100
42	90.3	24.7	4090
44	89.7	24.7	4085
46	89.2	24.7	4080
48	88.9	24.7	4075
50	88.5	24.7	4070
52	88.1	24.7	4065
54	87.7	24.7	4060
56	87.5	24.7	4050
58	87.2	24.7	4045
60	87.0	24.7	4040
62	86.8	24.7	4030
64	86.6	24.7	4010
66	86.4	24.7	3990
68	86.2	24.7	3970
70	86.0	24.7	3955
72	85.8	24.7	3945
74	85.6	24.7	3940
76	85.4	24.7	3930
78	85.2	24.7	3920
80	85.0	24.7	3910
82	84.8	24.8	3900
84	84.6	24.8	3890
86	84.4	24.8	3880
88	84.2	24.8	3870
90	84.0	24.8	3860
92	83.8	24.8	3850
94	83.6	24.8	3840
96	83.4	24.8	3830
98	83.2	24.8	3820
100	83.0	24.8	3810

EGG SIZE DISTRIBUTION—U.S. STANDARDS							
AGE (weeks)	AVERAGE EGG WT (lb / case)	% JUMBO Over 30 oz. / dozen	% EXTRA LARGE 27–30 oz. / dozen	% LARGE 24–27 oz. / dozen	% MEDIUM 21–24 oz. / dozen	% SMALL 18–21 oz. / dozen	% PEEWEE Under 18 oz. / dozen
20	38.0	0.00	0.02	2.92	32.09	54.42	10.55
22	41.3	0.00	0.42	13.72	57.05	27.29	1.52
24	43.6	0.03	2.78	32.09	52.46	12.20	0.44
26	45.0	0.18	6.12	43.68	43.32	6.54	0.16
28	45.9	0.24	9.87	48.76	37.40	3.68	0.05
30	46.5	0.40	12.67	52.82	31.48	2.61	0.01
32	47.0	0.43	15.09	56.01	26.96	1.50	0.01
34	47.4	0.61	16.51	58.14	23.62	1.12	0.00
36	47.7	0.57	18.84	59.19	20.70	0.70	0.00
38	47.9	0.60	19.89	60.17	18.65	0.69	0.00
40	48.1	0.79	22.04	59.10	17.47	0.60	0.00
42	48.3	1.02	23.43	59.22	15.74	0.59	0.00
44	48.4	1.09	24.25	58.53	15.59	0.54	0.00
46	48.6	1.50	26.88	56.26	14.82	0.54	0.00
48	48.7	1.74	27.96	55.10	14.67	0.54	0.00
50	48.8	1.87	28.63	54.38	14.63	0.49	0.00
52	48.9	2.09	29.95	53.54	13.94	0.47	0.00
54	49.0	2.56	30.61	53.36	13.01	0.45	0.00
56	49.1	2.79	31.14	52.71	12.93	0.43	0.00
58	49.2	2.97	31.92	51.78	12.91	0.42	0.00
60	49.3	3.98	33.64	49.27	12.69	0.42	0.00
62	49.4	4.02	33.83	49.21	12.53	0.40	0.00
64	49.5	4.16	34.62	49.11	11.73	0.38	0.00
66	49.6	4.23	35.05	48.88	11.46	0.38	0.00
68	49.7	4.60	36.53	47.90	10.59	0.38	0.00
70	49.8	4.80	37.49	46.96	10.39	0.37	0.00
72	49.9	4.81	37.86	46.94	10.02	0.37	0.00
74	50.0	5.00	37.92	46.94	9.78	0.36	0.00
76	50.0	5.02	38.02	46.93	9.68	0.35	0.00
78	50.0	5.12	38.04	46.82	9.67	0.35	0.00
80	50.0	5.13	38.10	46.77	9.66	0.33	0.00
82	50.0	5.23	38.23	46.66	9.56	0.32	0.00
84	50.0	5.23	38.37	46.63	9.47	0.29	0.00
86	50.0	5.26	38.38	46.61	9.46	0.28	0.00
88	50.1	5.26	38.41	46.60	9.46	0.26	0.00
90	50.1	5.29	38.49	46.58	9.38	0.26	0.00
92	50.1	5.33	38.54	46.57	9.31	0.24	0.00
94	50.1	5.41	38.79	46.56	9.01	0.22	0.00
96	50.1	5.41	39.14	46.49	8.74	0.21	0.00
98	50.1	5.46	39.19	46.48	8.69	0.18	0.00
100	50.1	5.46	39.49	46.48	8.41	0.16	0.00

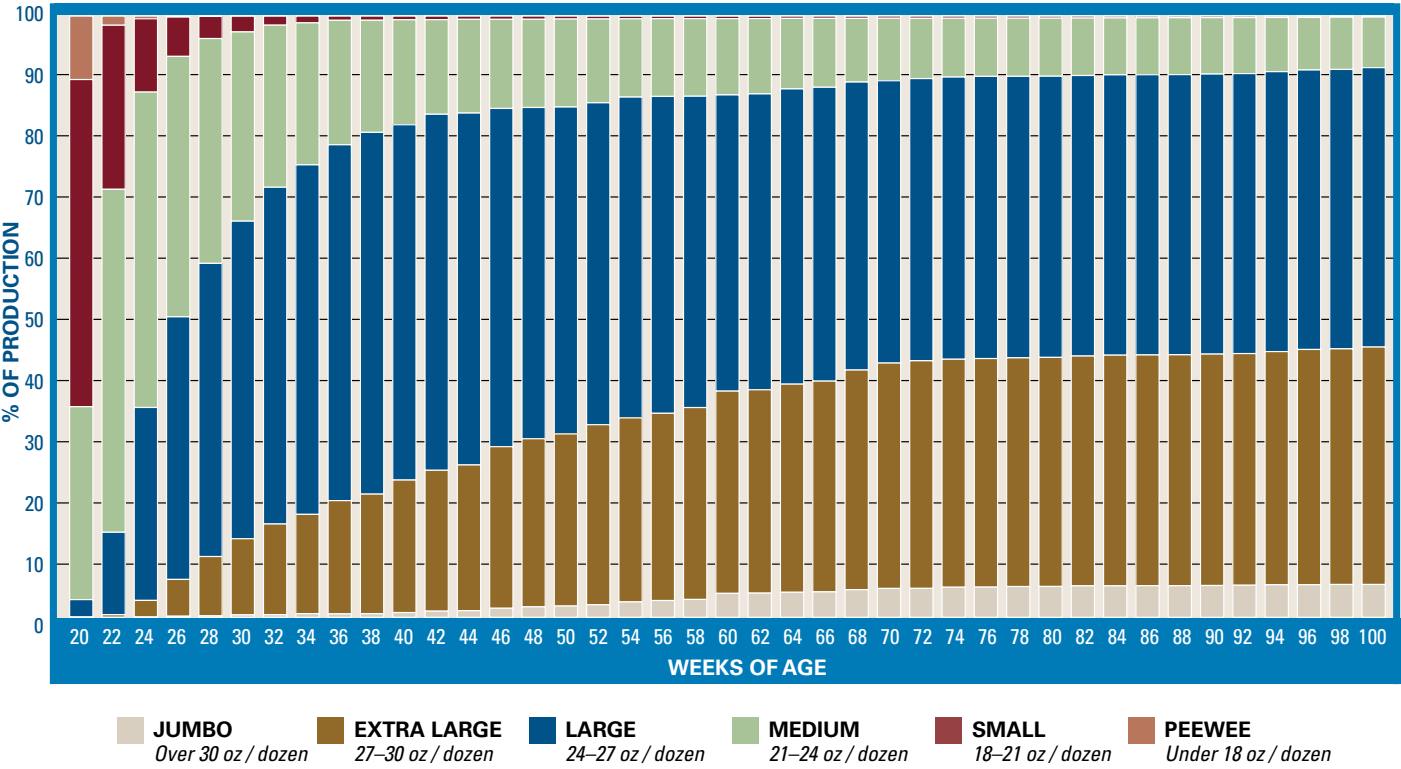
SHELL COLOR SCORES



- The W-36 normally lays uniformly white colored eggs (05–10).
- Eggs laid the first week are typically darker (10–20) than subsequent eggs.
- For more information on egg quality, see the [“The Science of Egg Quality”](#) technical update at www.hylinena.com.

Egg Size Distribution *(continued)*

EGG SIZE DISTRIBUTION—U.S. STANDARDS



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Feed Ingredient Table 1

INGREDIENT (as-fed basis)	DRY MATTER (%)	CRUDE PROTEIN (%)	FAT—ether extract (%)	CRUDE FIBER (%)	CALCIUM (%)	PHOSPHORUS total (%)	PHOSPHORUS available (%)	SODIUM (%)	CHLORIDE (%)	POTASSIUM (%)	SULFUR (%)	ME (kcal/lb)	ME (kcal/kg)	ME (MJ/kg)	LINOLEIC ACID (%)	CHOLINE (mg/kg)
Barley, grain	89.0	11.5	1.9	5.0	0.08	0.42	0.15	0.03	0.14	0.56	0.15	1250	2750	11.51	1.1	1027
Beans, broad (vicia faba)	89.0	25.7	1.4	8.2	0.14	0.54	0.20	0.08	0.04	1.20	—	1100	2420	10.13	0.9	1670
Calcium carbonate (38%Ca)	99.0	—	—	—	38.00	—	—	0.06	—	0.06	—	—	—	—	—	—
Canola meal (38%)	91.0	38.0	3.8	11.1	0.68	1.20	0.40	—	—	1.29	1.00	960	2110	8.83	—	6700
Corn, yellow, grain	86.0	7.5	3.5	1.9	0.01	0.28	0.12	0.02	0.04	0.33	0.08	1530	3373	14.11	1.9	1100
Corn gluten meal (60%)	90.0	60.0	2.0	2.5	0.02	0.50	0.18	0.03	0.05	0.45	0.50	1700	3740	15.65	1.8	2200
Cottonseed meal (41%), mech. extd	91.0	41.0	3.9	12.6	0.17	0.97	0.32	0.04	0.04	1.20	0.40	955	2100	8.79	0.8	2807
Cottonseed meal (41%), direct solv.	90.0	41.0	2.1	11.3	0.16	1.00	0.32	0.04	0.04	1.16	0.30	915	2010	8.41	0.4	2706
Dicalcium phosphate (18.5% P)	99.0	—	—	—	22.00	18.50	18.50	0.08	—	0.07	—	—	—	—	—	—
DL-Methionine	99.0	58.1	—	—	—	—	—	—	—	—	—	2277	5020	21.00	—	—
Fat, animal	99.0	—	98.0	—	—	—	—	—	—	—	—	3600	7920	33.14	—	—
Fat, vegetable	99.0	—	99.0	—	—	—	—	—	—	—	—	4000	8800	36.82	40.0	—
Fish meal, anchovy, Peruvian	91.0	65.0	10.0	1.0	—	—	—	0.88	0.60	0.90	0.54	1280	2820	11.80	0.1	5100
Fish meal, white	91.0	61.0	4.0	1.0	—	—	—	0.97	0.50	1.10	0.22	1180	2600	10.88	0.1	4050
Flaxseed	92.0	22.0	34.0	6.5	—	—	—	0.08	—	1.50	—	1795	3957	16.56	54.0	3150
L-Lysine	99.0	93.4	—	—	—	—	—	—	—	—	—	1868	4120	17.24	—	—
L-Threonine	99.0	72.4	—	—	—	—	—	—	—	—	—	1619	3570	14.94	—	—
L-Tryptophan	99.0	84.0	—	—	—	—	—	—	—	—	—	2653	5850	24.48	—	—
Linseed meal flax, expeller	90.0	32.0	3.5	9.5	0.40	0.80	—	0.11	—	1.24	0.39	700	1540	6.44	0.5	672
Linseed meal flax, solvent	88.0	33.0	0.5	9.5	0.35	0.75	—	0.14	—	1.38	0.39	635	1400	5.86	0.1	1760
Meat and bone meal, 50%	93.0	50.0	8.5	2.8	9.20	4.70	4.70	0.80	0.75	1.40	0.40	1150	2530	10.59	0.5	2000
Millet, pearl grain	90.0	12.0	4.2	1.8	0.05	0.30	0.10	0.04	0.64	0.43	0.13	1470	3240	13.56	1.3	789
Mono-dicalcium phosphate (21% P)	99.0	—	—	—	16.00	21.00	—	0.05	—	0.06	—	—	—	—	—	—
Oats, grain	90.0	11.0	4.0	10.5	0.10	0.35	0.14	0.07	0.12	0.37	0.21	1160	2550	10.67	2.4	1070
Peanut meal, solvent	90.0	47.0	2.5	8.4	0.08	0.57	0.18	0.07	0.03	1.22	0.30	1217	2677	11.20	0.5	1948
Poultry byproduct meal (feed grade)	94.0	57.0	14.0	2.5	5.00	2.70	2.70	0.30	0.55	0.60	0.50	1406	3100	12.97	0.7	5980
Rice bran, unextracted	91.0	13.5	5.9	13.0	0.10	1.70	0.24	0.10	0.07	1.35	0.18	925	2040	8.54	5.2	1948
Rice, grain, rough	89.0	7.3	1.7	10.0	0.04	0.26	0.09	0.04	0.06	0.34	0.10	1335	2940	12.30	0.83	5980
Safflower seed meal, expeller	91.0	20.0	6.6	32.2	0.23	0.61	0.20	0.05	0.16	0.72	0.10	525	1160	4.85	—	800
Salt, NaCl	99.0	—	—	—	—	—	—	39.34	60.66	—	—	—	—	—	—	—
Sodium bicarbonate, NaHCO ₃	99.0	—	—	—	—	—	—	27.38	—	—	—	—	—	—	—	—
Sorghum, milo, grain	89.0	11.0	2.8	2.0	0.04	0.29	0.10	0.03	0.09	0.34	0.09	1505	3310	13.85	1.3	678
Soybeans, full-fat, cooked	90.0	38.0	18.0	5.0	0.25	0.59	0.20	0.04	0.03	1.70	0.30	1520	3350	14.02	9.9	2420
Soybean meal, expeller	89.0	42.0	3.5	6.5	0.20	0.60	0.20	0.04	0.02	1.71	0.33	1100	2420	10.13	1.8	2673
Soybean meal, solvent	90.0	44.0	0.5	7.0	0.25	0.60	0.20	0.04	0.02	1.97	0.43	1020	2240	9.37	0.3	2743
Sunflower meal, expeller	93.0	41.0	7.6	21.0	0.43	1.00	0.25	0.20	0.01	1.00	0.10	1050	2310	9.67	6.5	—
Sunflower meal, partially dehul, solv.	92.0	34.0	0.5	13.0	0.30	1.25	0.27	0.20	0.01	1.60	0.38	1025	2260	9.46	0.2	1909
Triticale	90.0	12.5	1.5	2.59	0.05	0.30	0.10	—	0.07	—	0.20	1430	3150	13.18	0.9	460
Wheat, hard grain	88.0	13.5	1.9	3.0	0.05	0.41	0.12	0.06	0.07	0.50	0.10	1440	3170	13.26	1.00	778
Wheat, soft grain	86.0	10.8	1.7	2.8	0.05	0.30	0.11	0.06	0.07	0.40	0.10	1460	3210	13.43	1.00	778
Wheat bran	89.0	14.8	4.0	10.0	0.14	1.17	0.38	0.06	0.14	1.20	0.22	590	1300	5.44	2.10	980
Wheat middlings	89.0	15.0	3.6	8.5	0.15	1.17	0.45	0.06	0.07	0.60	0.16	950	2090	8.74	1.90	110

Nutrient recommendations are based on calculations using these energy and nutrient values (source: 2018–2019 Feedstuffs Reference Issue and field data). Values provided are “typical” based on ingredient surveys. Nutrient values should be confirmed by analysis of the materials being used in order to maintain an accurate formulation matrix.

Feed Ingredient Table 2

INGREDIENT (as-fed basis)	CRUDE PROTEIN (%)	LYSINE (%)		METHIONINE (%)		CYSTINE (%)		THREONINE (%)		TRYPTOPHAN (%)		ARGININE (%)		ISOLEUCINE (%)		VALINE (%)	
		Total content	Digestible content	Total content	Digestible content	Total content	Digestible content	Total content	Digestible content	Total content	Digestible content	Total content	Digestible content	Total content	Digestible content	Total content	Digestible content
Barley, grain	11.50	0.53	0.41	0.18	0.14	0.25	0.20	0.36	0.28	0.17	0.12	0.50	0.43	0.42	0.34	0.62	0.50
Beans, broad (vicia faba)	25.70	1.52	1.29	0.25	0.18	0.14	0.09	0.98	0.77	0.24	0.16	2.20	1.91	1.00	0.73	1.22	0.88
Canola meal (38%) ¹	91.0	2.02	1.60	0.77	0.69	0.97	0.71	1.50	1.17	0.46	0.38	2.30	2.07	1.51	1.25	1.94	1.59
Corn, yellow, grain	7.50	0.24	0.19	0.18	0.16	0.18	0.15	0.29	0.24	0.07	0.06	0.40	0.36	0.29	0.26	0.42	0.37
Corn gluten meal (60%)	60.00	1.00	0.88	1.90	1.84	1.10	0.95	2.00	1.84	0.30	0.25	1.90	1.82	2.30	2.19	2.70	2.57
Cottonseed meal (41%), mech. extd	41.00	1.52	0.99	0.55	0.40	0.59	0.44	1.30	0.88	0.50	0.39	4.33	3.81	1.31	0.93	1.84	1.36
Cottonseed meal (41%), direct solv.	41.00	1.70	1.11	0.51	0.37	0.62	0.46	1.31	0.89	0.52	0.41	4.66	4.10	1.33	0.95	1.82	1.34
DL-Methionine	58.10	—	—	99.00	99.00	—	—	—	—	—	—	—	—	—	—	—	—
Fish meal, anchovy, Peruvian	65.00	4.90	4.21	1.90	1.63	0.60	0.43	2.70	2.17	0.75	0.59	3.38	2.77	3.00	2.55	3.40	2.82
Fish meal, white	61.00	4.30	3.70	1.65	1.42	0.75	0.54	2.60	2.09	0.70	0.55	4.20	3.44	3.10	2.64	3.25	2.70
Flaxseed	22.00	0.92	0.79	0.35	0.30	0.42	0.30	0.77	0.62	0.22	0.17	2.05	1.68	0.95	0.81	1.17	0.97
L-Lysine	93.40	78.80	78.80	—	—	—	—	—	—	—	—	—	—	—	—	—	—
L-Threonine	72.40	—	—	—	—	—	—	98.50	98.50	—	—	—	—	—	—	—	—
L-Tryptophan	84.00	—	—	—	—	—	—	—	—	98.00	98.00	—	—	—	—	—	—
Linseed meal flax, expeller	32.00	1.10	0.99	0.47	0.37	0.56	0.44	1.10	1.00	0.47	0.43	2.60	2.39	1.70	1.49	1.50	1.29
Linseed meal flax, solvent	33.00	1.10	0.99	0.48	0.38	0.58	0.45	1.20	1.10	0.48	0.44	2.70	2.48	1.80	1.58	1.60	1.38
Meat and bone meal, 50%	50.00	2.60	2.05	0.67	0.57	0.33	0.19	1.70	1.34	0.26	0.13	3.35	2.85	1.70	1.41	2.25	1.85
Millet, pearl grain	12.00	0.35	0.32	0.28	0.25	0.24	0.20	0.44	0.37	0.20	0.18	0.55	0.49	0.52	0.46	0.70	0.62
Oats, grain	11.00	0.40	0.35	0.20	0.17	0.21	0.18	0.28	0.24	0.18	0.14	0.80	0.75	0.53	0.47	0.62	0.55
Peanut meal, solvent	47.00	1.52	1.29	0.50	0.44	0.60	0.47	1.12	0.91	0.42	0.39	4.76	4.28	1.50	1.32	1.80	1.57
Poultry byproduct meal (feed grade)	57.00	2.25	1.80	0.91	0.78	0.90	0.55	1.88	1.50	0.50	0.26	3.50	3.08	2.10	1.79	2.32	1.93
Rice bran, unextracted	13.50	0.50	0.38	0.17	0.13	0.10	0.07	0.40	0.28	0.10	0.08	0.45	0.39	0.39	0.30	0.60	0.46
Rice, grain, rough	7.30	0.24	0.19	0.14	0.13	0.08	0.07	0.27	0.22	0.12	0.11	0.59	0.54	0.33	0.27	0.46	0.39
Safflower seed meal, expeller	20.00	0.70	0.58	0.40	0.35	0.58	0.45	0.47	0.34	0.30	0.24	1.20	1.01	0.28	0.22	1.00	0.87
Sorghum, milo, grain	11.00	0.27	0.21	0.10	0.09	0.20	0.17	0.27	0.22	0.09	0.08	0.40	0.30	0.60	0.53	0.53	0.46
Soybeans, full-fat, cooked	38.00	2.40	2.16	0.54	0.49	0.55	0.45	1.69	1.43	0.52	0.46	2.80	2.60	2.18	1.94	2.02	1.78
Soybean meal, expeller	42.00	2.70	2.43	0.60	0.54	0.62	0.51	1.70	1.44	0.58	0.52	3.20	2.97	2.80	2.49	2.20	1.94
Soybean meal, solvent	44.00	2.70	2.43	0.65	0.58	0.67	0.55	1.70	1.44	0.60	0.53	3.40	3.16	2.50	2.22	2.40	2.11
Sunflower meal, expeller	41.00	2.00	1.74	1.60	1.47	0.80	0.64	1.60	1.31	0.60	0.52	4.20	3.91	2.40	2.14	2.40	2.08
Sunflower meal, partially dehul, solv.	34.00	1.42	1.19	0.64	0.60	0.55	0.43	1.48	1.26	0.35	0.30	2.80	2.32	1.39	1.25	1.64	1.41
Triticale	12.50	0.39	0.35	0.26	0.23	0.26	0.22	0.36	0.31	0.14	0.12	0.48	0.39	0.76	0.70	0.51	0.44
Wheat, hard grain	13.50	0.40	0.32	0.25	0.22	0.30	0.26	0.35	0.29	0.18	0.16	0.60	0.53	0.69	0.61	0.69	0.59
Wheat, soft grain	10.80	0.30	0.24	0.14	0.12	0.20	0.17	0.28	0.23	0.12	0.11	0.40	0.35	0.43	0.38	0.48	0.41
Wheat bran	14.80	0.60	0.43	0.20	0.15	0.30	0.22	0.48	0.35	0.30	0.24	1.07	0.88	0.60	0.47	0.70	0.54
Wheat Middlings	15.00	0.70	0.56	0.12	0.10	0.19	0.14	0.50	0.36	0.20	0.16	1.00	0.80	0.70	0.58	0.80	0.61

Amino acid digestibility is standardized ileal digestibility. Amino acid values are standardized for 88% dry matter (source: 2018–2019 Feedstuffs Reference Issue and field data). Values provided are “typical” based on ingredient surveys. Nutrient values should be confirmed by analysis of the materials being used in order to maintain an accurate formulation matrix.

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Hy-Line North America Welfare Goals and Principles

To promote animal well-being and produce birds of the highest quality, we adhere to the following welfare goals and principles. These goals and principles are the essential building blocks for the humane and professional care of our birds:

- **Feed and Water**
Provide access to good quality water and nutritionally balanced diets at all times
- **Health and Veterinary Care**
Provide science-based health programs and prompt veterinary care
- **Environment**
Provide shelter that is designed, maintained and operated to meet the bird's needs and to facilitate daily inspection
- **Husbandry and Handling Practices**
Provide comprehensive care and handling procedures that ensure the bird's well-being throughout its life
- **Transportation**
Provide transportation that minimizes travel time and stress

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